

CS1231S · Condensed Revision

Lectures 1–6, Tutorials 1–5 and the AY23–25 midterms. Definitions, theorems and worked examples are integrated by topic; Appendix A is complete. Theorem and appendix labels follow the lecture slides.

Find material: topic locator §32; theorem locator §34; past-paper/tutorial locator §29. **Test format (past papers):** 90 minutes, open book, 50 marks: 30 MCQ + 20 written.

Conventions: $\mathbb{N} = \{0, 1, 2, \dots\}$; $\bar{A}$ is complement; $A \setminus B$ is difference; use $\subseteq$ for subset. Largest/smallest = greatest/least. Blue boxes define terms; grey boxes contain proofs and worked examples; green boxes state results; amber boxes solve past papers.

Part I · Propositional Logic

1 · Statements, connectives, truth tables

DEFINITION Statement (2.1.1) · Statement form (2.1.5)

A **statement** (proposition) is a sentence that is true or false, but not both. A **statement form** is an expression built from statement variables and connectives that becomes a statement when statements are substituted for its variables.

DEFINITION Negation (2.1.2) · Conjunction (2.1.3) · Disjunction (2.1.4)

$\sim p$: “not p ”; $p \wedge q$: “ p and q ”; $p \vee q$: inclusive “ p or q ”, true when at least one is true.

p	q	$\sim p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

Order of operations. $\sim$ first; then $\wedge$ and $\vee$, which are **coequal**, so $p \wedge q \vee r$ is ambiguous and must be bracketed; then $\rightarrow, \leftrightarrow$ last. (CS2100 puts $\wedge$ before $\vee$; CS1231S does not.)

Exclusive or: $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$.

$A \rightarrow$ is false in exactly one row: **T** $\rightarrow$ **F**.

DEFINITION Tautology, contradiction (2.1.7–8) · Logical equivalence (2.1.6)

A **tautology** is a statement form that is true for every assignment; a **contradiction** is false for every assignment. A statement whose form is a tautology (contradiction) is a **tautological (contradictory)** statement. $P \equiv Q$ iff P and Q have identical truth values under every assignment; equivalently, iff $P \leftrightarrow Q$ is a tautology.

2 · Logical equivalences and set identities

Theorem 2.1.1 (logic) and Theorem 6.2.2 (sets) side by side (Lecture 2, slides 25–26; Lecture 5, slides 42–43). Read $\wedge$ as $\cap$, $\vee$ as $\cup$, $\sim$ as complement, t as U , c as $\emptyset$. Cite laws **by name**.

Law	Logic (for statement variables)	Sets (subsets of U)
Commutative	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$	$A \cap B = B \cap A$ $A \cup B = B \cup A$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(A \cap B) \cap C = A \cap (B \cap C)$ $(A \cup B) \cup C = A \cup (B \cup C)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$p \wedge t \equiv p$ $p \vee c \equiv p$	$A \cap U = A$ $A \cup \emptyset = A$
Negation / Complement	$p \vee \sim p \equiv t$ $p \wedge \sim p \equiv c$	$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$
Double negation / complement	$\sim(\sim p) \equiv p$	$\overline{\bar{A}} = A$
Idempotent	$p \wedge p \equiv p$ $p \vee p \equiv p$	$A \cap A = A$ $A \cup A = A$
Universal bound	$p \vee t \equiv t$ $p \wedge c \equiv c$	$A \cup U = U$ $A \cap \emptyset = \emptyset$
De Morgan's	$\sim(p \wedge q) \equiv \sim p \vee \sim q$ $\sim(p \vee q) \equiv \sim p \wedge \sim q$	$\overline{A \cap B} = \bar{A} \cup \bar{B}$ $\overline{A \cup B} = \bar{A} \cap \bar{B}$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$
Negations of t, c	$\sim t \equiv c$ $\sim c \equiv t$	$\bar{U} = \emptyset$ $\bar{\emptyset} = U$
Set difference	—	$A \setminus B = A \cap \bar{B}$

THEOREM Further equivalences

- **Implication law:** $p \rightarrow q \equiv \sim p \vee q$
- **Contrapositive:** $p \rightarrow q \equiv \sim q \rightarrow \sim p$
- **Negated conditional:** $\sim(p \rightarrow q) \equiv p \wedge \sim q$
- **Biconditional:** $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- **Variant absorption:** $p \wedge (\sim p \vee q) \equiv p \wedge q$
- **Variant absorption:** $p \vee (\sim p \wedge q) \equiv p \vee q$
- **Exportation:** $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$
- **Cases:** $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$
- $p \rightarrow (q \vee r) \equiv (p \wedge \sim q) \rightarrow r$

The left column is proved in Lecture 2. The right column comes from tutorials and assignments; each follows from the table in a line or two, so derive it if a question says “use the laws” (the AY24 answer key cites variant absorption as an assignment result).

PAST PAPER AY24 Q4 · Simplify $((p \rightarrow q) \rightarrow (q \rightarrow r)) \rightarrow (r \rightarrow s) \vee p) \wedge (p \rightarrow (p \rightarrow q)) \wedge p$

Call the long first conjunct $X \vee p$. Answer: $p \wedge q$ (option C).

$$\begin{aligned}
 & (X \vee p) \wedge (p \rightarrow (p \rightarrow q)) \wedge p \\
 \equiv & (X \vee p) \wedge (p \wedge (p \rightarrow (p \rightarrow q))) && \text{(associative, commutative)} \\
 \equiv & ((X \vee p) \wedge p) \wedge (p \rightarrow (p \rightarrow q)) && \text{(associative)} \\
 \equiv & p \wedge (p \rightarrow (p \rightarrow q)) && \text{(commutative } \times 2, \text{ absorption)} \\
 \equiv & p \wedge (\sim p \vee (\sim p \vee q)) && \text{(implication law } \times 2) \\
 \equiv & p \wedge (\sim p \vee q) && \text{(associative, idempotent)} \\
 \equiv & p \wedge q && \text{(variant absorption)}
 \end{aligned}$$

METHOD Checking tautologies and equivalences

To test whether $P \rightarrow Q$ is a tautology, try to make it false: set Q false and P true, and follow the consequences. If this forces a contradiction, $P \rightarrow Q$ is a tautology. **AY24 Q3(iii)**

$(p \vee q \vee \sim r) \wedge (s \vee \sim(p \vee q)) \rightarrow (s \vee \sim r)$: falsifying needs $s = F, r = T$; the premise becomes $(p \vee q) \wedge \sim(p \vee q)$, a contradiction, so the statement is a tautology. **To show $P \not\equiv Q$** , one row is enough. AY23 Q3(i): $\sim(p \rightarrow q) \equiv p \vee (p \rightarrow \sim q)$? At $p = q = T$ the left side is F and the right side is T, so they are not equivalent.

WORKED EXAMPLE Logic laws

Without a truth table, show $(p \rightarrow q) \wedge (p \rightarrow \sim q) \equiv \sim p$.

$$\begin{aligned}
 & (p \rightarrow q) \wedge (p \rightarrow \sim q) \\
 \equiv & (\sim p \vee q) \wedge (\sim p \vee \sim q) && \text{(implication law } \times 2) \\
 \equiv & \sim p \vee (q \wedge \sim q) && \text{(distributive law)} \\
 \equiv & \sim p \vee \mathbf{c} && \text{(negation law)} \\
 \equiv & \sim p && \text{(identity law)}.
 \end{aligned}$$

■

3 · Conditional statements**DEFINITION Conditional (2.2.1) and its relatives (2.2.2–4)**

$p \rightarrow q$ (“if p then q ”, “ p implies q ”) is false only when p is true and q is false; when p is false it is **vacuously true**. p is the **hypothesis** (antecedent) and q the **conclusion** (consequent). From $p \rightarrow q$ form the **contrapositive** $\sim q \rightarrow \sim p$, the **converse** $q \rightarrow p$, and the **inverse** $\sim p \rightarrow \sim q$.

THEOREM

A conditional $\equiv$ its contrapositive. Its converse $\equiv$ its inverse (they are contrapositives of each other). A conditional is **not** equivalent to its converse or its inverse.

DEFINITION Only if (2.2.5) · Biconditional (2.2.6) · Necessary/sufficient (2.2.7)

The table below translates each phrase. “ p iff q ” is true exactly when their truth values agree. A necessary condition must hold; a sufficient condition guarantees the conclusion.

English	Symbolic	English	Symbolic
if p then q ; p implies q	$p \rightarrow q$	p only if q	$p \rightarrow q$
q if p ; q whenever p	$p \rightarrow q$	p unless q	$\sim q \rightarrow p \equiv p \vee q$
p is sufficient for q	$p \rightarrow q$	p is necessary for q	$q \rightarrow p$
p if and only if q	$p \leftrightarrow q$	neither p nor q	$\sim p \wedge \sim q$

TRAP Everyday “if” is often an “iff”

“If you behave, you will get ice-cream” is usually understood as a biconditional. In mathematics it is only $b \rightarrow i$; misbehaving and still getting ice-cream does not make it false.

4 · Arguments and rules of inference**DEFINITION** Argument, validity (2.3.1) · Critical row · Syllogism · Sound, unsound (2.3.2)

An **argument** (**argument form**) is a sequence of statements (statement forms); all but the last are **premises** (assumptions, hypotheses), the last (after $\therefore$) is the **conclusion**. An argument form is **valid** iff, whatever statements are substituted for its variables, if the premises are all true then the conclusion is true. A **critical row** is a truth-table row in which all premises are T: the form is valid iff the conclusion is T in every critical row. A **syllogism** is an argument form with two premises and a conclusion (e.g. modus ponens). An argument is **sound** iff it is valid **and** all its premises are true; otherwise it is **unsound**.

Rule	Form	Rule	Form
Modus ponens	$p \rightarrow q, p \therefore q$	Generalization	$p \therefore p \vee q$
Modus tollens	$p \rightarrow q, \sim q \therefore \sim p$	Specialization	$p \wedge q \therefore p$
Conjunction	$p, q \therefore p \wedge q$	Elimination	$p \vee q, \sim q \therefore p$
Transitivity	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$	Contradiction rule	$\sim p \rightarrow c \therefore p$
Division into cases	$p \vee q, p \rightarrow r, q \rightarrow r \therefore r$		

Fallacies (invalid): *converse error* $p \rightarrow q, q \therefore p$; *inverse error* $p \rightarrow q, \sim p \therefore \sim q$.

PAST PAPER AY24 Q5 · Which conclusions follow from $p \rightarrow (q \wedge r)$, $\sim p \rightarrow \sim r$, $\sim p \wedge (r \rightarrow \sim p)$?

- $\sim p$ (specialization, premise 3) then $\sim r$ (modus ponens with premise 2).
- So every critical row has $p = r = F$ (q free). Then (i) $p \rightarrow q$ is T (vacuous); (ii) $\sim p \vee \sim r$ is T; (iii) $q \rightarrow (p \rightarrow r)$ is T since $p \rightarrow r$ is T; (iv) $\sim p \wedge \sim q \wedge r$ is F since $r = F$.

Valid: (i), (ii), (iii). Answer C.

PAST PAPER AY25 Q13 · Sound vs valid

Sound $\Rightarrow$ valid by definition, so “ $\forall x(\text{Sound}(x) \rightarrow \text{Valid}(x))$ ” is **true**; the converse fails (valid with a false premise), so “ $\exists x(\sim \text{Sound}(x) \wedge \text{Valid}(x))$ ” is **true**; the other two are false. Answer: **only I and IV**.

PAST PAPER AY23 Q4 · Knights and knaves

Method: let A, D, N mean “is a knight”. A statement S made by X gives $X \leftrightarrow S$. Here: $A \leftrightarrow D$; $D \leftrightarrow \sim N$; “Aiken is not my type” gives $N \leftrightarrow \sim(A \leftrightarrow N)$; “Aiken lies” gives $N \leftrightarrow \sim A$. From $N \leftrightarrow \sim A$, A and N differ, so $\sim(A \leftrightarrow N)$ is T, hence N is T. Then $A = F$, $D = A = F$, and $D \leftrightarrow \sim N$ checks out. **Aiken and Dueet are knaves; Noasu is a knight.** Answer B.

PAST PAPER AY25 Q16 (6 marks) · Four animals, four colours, nine clues

Names: cat (3 letters), sheep (5), rabbit (6), penguin (7).

- (2) six letters $\leftrightarrow$ black: **rabbit** is black; cat, sheep, penguin are not.
- (3) an “e” $\rightarrow$ white: **sheep** and **penguin** are white.
- (4) a repeated letter $\rightarrow$ not grey: sheep (ee), rabbit (bb), penguin (nn) are not grey. (5) brown $\rightarrow$ has an “i”: cat and sheep are not brown.
- (7) sheep has exactly one colour: **sheep** = {white}. (9) someone is grey, and only the cat can be: **cat** = {grey} (exactly one, by (7)).
- (6) penguin has exactly two colours: white, and not black or grey, so **penguin** = {white, brown}.
- (8) some animal is exactly {brown, black}; only the rabbit is black: **rabbit** = {brown, black}. (1) holds.

Model derivations**PROOF** Simplify $\sim(\sim p \wedge q) \wedge (p \vee q)$ using the laws

$$\begin{aligned}
 &\sim(\sim p \wedge q) \wedge (p \vee q) \equiv (\sim\sim p \vee \sim q) \wedge (p \vee q) \\
 &\equiv (p \vee \sim q) \wedge (p \vee q) \\
 &\equiv p \vee (\sim q \wedge q) \\
 &\equiv p \quad \blacksquare
 \end{aligned}$$

(De Morgan’s law)

(double negative law)

(distributive law)

(commutative, negation and identity laws)

PROOF Derive variant absorption $p \wedge (\sim p \vee q) \equiv p \wedge q$ from the table

$$\begin{aligned} & p \wedge (\sim p \vee q) \\ & \equiv (p \wedge \sim p) \vee (p \wedge q) \quad (\text{distributive}) \\ & \equiv \mathbf{c} \vee (p \wedge q) \quad (\text{negation}) \\ & \equiv (p \wedge q) \vee \mathbf{c} \quad (\text{commutative}) \\ & \equiv p \wedge q \quad (\text{identity}). \end{aligned}$$

■

PROOF A deduction with named rules: from $p \vee q$, $q \rightarrow r$, $(p \wedge s) \rightarrow t$, $\sim r$, $\sim q \rightarrow (u \wedge s)$, conclude t

1. $\sim q$ (from $q \rightarrow r$ and $\sim r$, by *modus tollens*)
2. $u \wedge s$ (from $\sim q \rightarrow (u \wedge s)$ and step 1, by *modus ponens*) so s (*specialization*)
3. p (from $p \vee q$ and step 1, by *elimination*)
4. $p \wedge s$ (steps 2, 3, by *conjunction*) so t (from $(p \wedge s) \rightarrow t$, by *modus ponens*).

■

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

METHOD Validity by critical rows

For $p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$ (transitivity): the **critical rows** are those where every premise is T: rows 1, 5, 7 and 8 (bold). The conclusion is T in each, so the argument is **valid**. A critical row with a false conclusion would make the argument invalid; that row is a counterexample.

TRAP $\rightarrow$ is not associative

$(p \rightarrow q) \rightarrow r \not\equiv p \rightarrow (q \rightarrow r)$: at $p = q = r = F$ the left is $T \rightarrow F = F$, the right is T . Always bracket chains of $\rightarrow$.

WORKED EXAMPLE Validity

Decide validity: (a) $p \rightarrow q, q \rightarrow r, \sim r \therefore \sim p$. (b) $p \rightarrow q, \sim p \therefore \sim q$.

(a) **Valid**: $p \rightarrow r$ (transitivity) with $\sim r, \sim p$ (*modus tollens*).

(b) **Invalid** (inverse error): the critical row $p = F, q = T$ makes both premises true and the conclusion $\sim q$ false.

Part II · Quantified Statements

5 · Predicates and quantifiers

DEFINITION Predicate (3.1.1) · Truth set (3.1.2)

A **predicate** is a sentence with finitely many variables that becomes a statement when values are substituted. The **domain** of a variable is the set of values that may be substituted for it (also called the domain of discourse, universe of discourse, universal set, or universe). The **truth set** of $P(x)$, $x \in D$, is the set of all elements of D that make $P(x)$ true: $\{x \in D : P(x)\}$ (also written $\{x \in D \mid P(x)\}$).

DEFINITION Kinds of mathematical statements (Lecture 1)

Universal: “all/every/any” ($\forall$); **existential**: “some/at least one” ($\exists$); **conditional**: “if ... then” ($\rightarrow$).

Combined forms:

- **Universal conditional**: $\forall x (P(x) \rightarrow Q(x))$.
- **Universal existential**: $\forall x \exists y P(x, y)$.
- **Existential universal**: $\exists x \forall y P(x, y)$.

DEFINITION Universal (3.1.3) and existential (3.1.4) statements · Uniqueness

$\forall x \in D, Q(x)$ is true iff $Q(x)$ is true for **every** $x \in D$, and false iff $Q(x)$ is false for at least one x , called a **counterexample**.

$\exists x \in D$ such that $Q(x)$ is true iff $Q(x)$ is true for **at least one** $x \in D$, and false iff $Q(x)$ is false for all $x \in D$.

$\exists! x \in D, Q(x) \equiv \exists x \in D (Q(x) \wedge \forall y \in D (Q(y) \rightarrow y = x))$, read “there is exactly one”.

METHOD Translating English

Use $\rightarrow$ under $\forall$ and $\wedge$ under $\exists$. “All A are B ” is $\forall x(A(x) \rightarrow B(x))$; “some A is B ” is $\exists x(A(x) \wedge B(x))$. Writing $\forall x(A(x) \wedge B(x))$ claims **everything** is an A ; writing $\exists x(A(x) \rightarrow B(x))$ is true as soon as one non- A exists.

Implicit quantification: “If $x > 2$ then $x^2 > 4$ ” in a real-number context means $\forall x \in \mathbb{R}(x > 2 \rightarrow x^2 > 4)$.

PAST PAPER AY24 Q2 · “All birds cannot fly”

$\forall x(Bird(x) \rightarrow \sim Fly(x))$, option C. Option B, $\forall x(Bird(x) \wedge \sim Fly(x))$, asserts that everything is a bird; D quantifies a second, unrelated y .

6 · Negation, vacuous truth, variants**THEOREM Theorem 3.2.1 · Theorem 3.2.2 · Negation of quantified statements**

$\sim(\forall x \in D, P(x)) \equiv \exists x \in D, \sim P(x)$ $\sim(\exists x \in D, P(x)) \equiv \forall x \in D, \sim P(x)$

Universal conditional: $\sim\forall x \in D(P(x) \rightarrow Q(x)) \equiv \exists x \in D(P(x) \wedge \sim Q(x))$.

Nested: push $\sim$ inward, flipping each quantifier: $\sim\forall x\exists y P(x, y) \equiv \exists x\forall y\sim P(x, y)$.

DEFINITION Vacuous truth

$\forall x \in D(P(x) \rightarrow Q(x))$ is **vacuously true** when no $x \in D$ satisfies $P(x)$. In particular, every universal statement over an **empty** domain is true, and every existential statement over an empty domain is false.

DEFINITION Variants (3.2.1) · Necessary, sufficient, only if (3.2.2)

For $\forall x \in D(P(x) \rightarrow Q(x))$: **contrapositive** $\forall x(\sim Q(x) \rightarrow \sim P(x))$ (equivalent), **converse**

$\forall x(Q(x) \rightarrow P(x))$, **inverse** $\forall x(\sim P(x) \rightarrow \sim Q(x))$.

“ $r(x)$ is sufficient for $s(x)$ ”: $\forall x(r(x) \rightarrow s(x))$. “ $r(x)$ is necessary for $s(x)$ ”: $\forall x(s(x) \rightarrow r(x))$. “ $r(x)$ only if $s(x)$ ”: $\forall x(r(x) \rightarrow s(x))$.

PAST PAPER AY23 Q2 · Given $\forall x \in S(SoC(x) \wedge CS1231(x))$ is true

(i) All computing students took CS1231S: **true**. (ii) Every student is a computing student: **true**, because of the $\wedge$. (iii) $S \neq \emptyset$: **false**, since the statement is vacuously true when $S = \emptyset$. (iv) Some non-computing student took it: **false**, as there is no such student. Only (i) and (ii) hold; no option lists this, so the answer is **E**.

PAST PAPER AY25 Q2 · $\forall x \in D(Red(x) \wedge Flying(x))$ for the objects in LT11

A: if the statement is true, every object is red. True. B: if true, every object flies. True. C: it is true when LT11 is empty. True, vacuously. D: if it is false, there is a counterexample object, so LT11 is non-empty. True. Options A–D are all correct, so the false option is **E**.

7 · Multiple quantifiers**THEOREM Order of quantifiers**

Adjacent quantifiers of the same kind commute: $\forall x\forall y \equiv \forall y\forall x$ and $\exists x\exists y \equiv \exists y\exists x$. Swapping a $\forall$ past an $\exists$ usually changes the meaning: in $\forall x\exists y P(x, y)$ the witness y **may depend on** x ; in $\exists y\forall x P(x, y)$ one y must work for all x . $\exists y\forall x P \Rightarrow \forall x\exists y P$, never conversely in general.

METHOD Evaluating and proving nested statements

Read the quantifiers from left to right. A $\forall$ variable may take any value; an $\exists$ variable is chosen afterwards and may depend on the values chosen before it.

To **prove** $\forall x\exists y P$: “Let x be arbitrary. Let $y = f(x)$. Then $P(x, f(x))$ because ...”. To **disprove** it, prove the negation $\exists x\forall y\sim P$: give one x and show that every y fails for it.

PAST PAPER AY23 Q5 · Domain $\mathbb{R}$

(i) $\forall x\exists y(xy = 1)$ is **false**: at $x = 0$, $0 \cdot y = 0 \neq 1$ for every y .

(ii) $\exists x\forall y(x^2 \geq y^2)$ is **false**: for any x , $y = |x| + 1$ gives $y^2 > x^2$.

(iii) $\exists x\forall y\exists z(xy^2 = yz)$ is **true**: take $x = 0$; for any y choose $z = 0$. (Or $x = 1$, $z = y$.) Answer **B**.

PAST PAPER AY23 Q16 (6 marks) · Solving for the witnesses

(a) $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$; smallest positive integer that can lie in C with

$\forall x \in A \forall y \in B \forall z \in C (|x - y| \leq |y - z|)$. For each y the worst x is 1, so we need $|y - z| \geq y - 1$ for $y = 4, 5, 6$: at $z = 1$, $|y - 1| = y - 1$. So $z = 1$.

(b) $\exists y \in \mathbb{Z}^+ \forall x \in \{10, \dots, 50\} \forall z \in \{3, 5, 7, 9\} (x \leq y^2 \wedge y \leq z^2)$: need $y^2 \geq 50$, so $y \geq 8$, and $y \leq 3^2 = 9$. So $y \in \{8, 9\}$.

(c) $\forall x \in \mathbb{Z} \exists y \in \mathbb{R} \setminus \{0\} (xy < 1)$: take $y = -1$ if $x \geq 0$, $y = 1$ if $x < 0$; then $xy = -x \leq 0$ or $xy = x < 0$, both < 1 . (The key lists several others, e.g. $y = 1$ at $x = 0$ and $y = -x$ otherwise.)

TRAP Hidden domain changes

$\forall x \in \mathbb{Z}^+ \exists y \in \mathbb{Z} (x + y = 0)$ is true but becomes false over $y \in \mathbb{Z}^+$. Check every domain again after negating or reordering quantifiers.

WORKED EXAMPLE Nested quantifiers

Decide and justify: (a) $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (y > x^2)$; (b) $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (y > x^2)$.

(a) **True.** Let $x \in \mathbb{Z}$ be arbitrary; take $y = x^2 + 1 \in \mathbb{Z}$ (closure) then $y > x^2$.

(b) **False.** Its negation $\forall y \in \mathbb{Z} \exists x \in \mathbb{Z} (y \leq x^2)$ holds: given y , take $x = y$; then $y \leq y^2$ for every integer y (if $y \leq 0$ then $y \leq 0 \leq y^2$; if $y \geq 1$ then $y^2 = y \cdot y \geq y$).

8 · Arguments with quantified statements**DEFINITION Valid argument form (3.4.1)**

Valid iff every substitution of predicates that makes all premises true also makes the conclusion true.

An argument is valid iff its form is valid.

Rule	Form	Rule	Form
Universal instantiation	$\forall x \in D P(x); a \in D \therefore P(a)$	Universal generalization	$P(a)$ for an arbitrary $a \in D \therefore \forall x \in D P(x)$
Universal modus ponens	$\forall x (P(x) \rightarrow Q(x)); P(a) \therefore Q(a)$	Universal modus tollens	$\forall x (P(x) \rightarrow Q(x)); \sim Q(a) \therefore \sim P(a)$
Existential instantiation	$\exists x P(x) \therefore P(c)$ for some new c	Existential generalization	$P(a); a \in D \therefore \exists x \in D P(x)$

Invalid: quantified **converse error** $\forall x (P(x) \rightarrow Q(x)); Q(a) \therefore P(a)$ and **inverse error** $\forall x (P(x) \rightarrow Q(x)); \sim P(a) \therefore \sim Q(a)$.

Validity with diagrams: draw “all P are Q ” as the P -disc inside the Q -disc; the argument is valid iff **every** consistent drawing forces the conclusion.

PAST PAPER AY25 Q6 (number theory in quantifier form) · n a perfect square

(i) “ n is prime” is false ($4 = 2^2$). (ii) “ n has an odd number of positive factors” is true for $n \geq 1$ (divisors pair up as $d \leftrightarrow n/d$, except $d = \sqrt{n}$), but **$n = 0$ is a perfect square with infinitely many factors**, so the key accepted both C and D. (iii) “ $p \mid n \Rightarrow p^2 \mid n$ ” is true: if $n = m^2$ and $p \mid m^2$ then $p \mid m$ (unique factorization), so $p^2 \mid m^2$.

Model solutions with nested quantifiers**PROOF Negating the definition of a limit (Lecture 1)**

$\lim_{n \rightarrow \infty} a_n = L$ means $\forall \varepsilon \in \mathbb{R}^+ \exists N \in \mathbb{Z} \forall n \in \mathbb{Z} (n > N \rightarrow L - \varepsilon < a_n < L + \varepsilon)$.

Its negation, one quantifier at a time: $\exists \varepsilon \in \mathbb{R}^+ \forall N \in \mathbb{Z} \exists n \in \mathbb{Z} (n > N \wedge (a_n \leq L - \varepsilon \vee a_n \geq L + \varepsilon))$. ■

PROOF $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (x + y = 0)$ is true; $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (x + y = 0)$ is false

1. Let x be an arbitrary integer and let $y = -x$. Then $y \in \mathbb{Z}$ (closure) and $x + y = 0$.

2. For the second, prove the negation $\forall y \in \mathbb{Z} \exists x \in \mathbb{Z} (x + y \neq 0)$: given any y , take $x = 1 - y$; then $x + y = 1 \neq 0$. ■

English	Symbolic ($L(x, y)$: x loves y)
Everybody loves somebody.	$\forall x \exists y L(x, y)$
Somebody is loved by everybody.	$\exists y \forall x L(x, y)$
Nobody loves everybody.	$\sim \exists x \forall y L(x, y) \equiv \forall x \exists y \sim L(x, y)$
There is exactly one person everybody loves.	$\exists! y \forall x L(x, y)$

PROOF Formal negation of a nested quantified conditional (Lecture 3, Tarski's world)

“There is a square x such that for all triangles y , x is to the right of y ”:

$\exists x (Square(x) \wedge \forall y (Triangle(y) \rightarrow RightOf(x, y)))$. Negate step by step:

$$\begin{aligned} &\sim \exists x (Square(x) \wedge \forall y (Triangle(y) \rightarrow RightOf(x, y))) \\ &\equiv \forall x \sim (Square(x) \wedge \forall y (Triangle(y) \rightarrow RightOf(x, y))) && \text{(negation of } \exists \text{)} \\ &\equiv \forall x (\sim Square(x) \vee \sim \forall y (Triangle(y) \rightarrow RightOf(x, y))) && \text{(De Morgan)} \\ &\equiv \forall x (\sim Square(x) \vee \exists y \sim (Triangle(y) \rightarrow RightOf(x, y))) && \text{(negation of } \forall \text{)} \\ &\equiv \forall x (\sim Square(x) \vee \exists y (Triangle(y) \wedge \sim RightOf(x, y))) && \text{(negated conditional)} \end{aligned}$$

“Every object is either not a square, or has some triangle it is not to the right of.” ■

DRILL Translate, then negate (answers after the arrow)

- “Every student has a friend”: $\forall s \exists f F(s, f) \rightarrow$ negation $\exists s \forall f \sim F(s, f)$.
- “No integer is both even and odd”: $\forall n \sim (E(n) \wedge O(n)) \rightarrow$ negation $\exists n (E(n) \wedge O(n))$.
- “All primes greater than 2 are odd”: $\forall p (P(p) \wedge p > 2 \rightarrow O(p)) \rightarrow \exists p (P(p) \wedge p > 2 \wedge \sim O(p))$.
- “ x is a **minimal** element”: $\forall y (y \preceq x \rightarrow y = x) \rightarrow \exists y (y \preceq x \wedge y \neq x)$.

Part III · Number Theory and Methods of Proof

9 · Writing proofs

Term	Meaning (Lecture 1)
Definition	precise meaning of a term: all and only the required properties
Axiom / postulate	assumed true without proof (Peano, Euclid's postulates)
Theorem	a major result, proved rigorously
Lemma	a small result used to prove a theorem
Corollary	a simple deduction from a theorem
Conjecture	believed true, unproved (Goldbach)

METHOD Proof format

- **Number the steps;** nest sub-arguments (1.1, 1.2, ...).
- **Justify every step** in brackets: “by definition of odd”, “by closure of integers under +”, “by basic algebra”, “by Theorem 4.3.3”.
- Open a universal proof with “Let x be a **arbitrary** element of D ”.
- Give each existential witness a **fresh** name.
- End with the claim, restated, then ■.

DEFINITION Basic properties of integers (Lecture 1, Appendix A)

For all $x, y, z \in \mathbb{Z}$: **closure** ($x + y, x - y, xy \in \mathbb{Z}$), **commutativity**, **associativity**, **distributivity**, **trichotomy** (exactly one of $x < y$, $x = y$, $x > y$). Appendix A's field and order axioms for $\mathbb{R}$ may also be cited; the complete Appendix A appears in §33.

Assumption 1: every integer is even or odd, but not both.

Assumption 2: every rational number

can be reduced to lowest terms.

TRAP Common mistakes in proofs

Arguing from examples · reusing a letter ($m = 2r$ and $n = 2r$ makes $m = n$) · assuming what is to be proved
 · “therefore” without a reason · writing “ $a \mid b = 3$ ” ($a \mid b$ is a statement, not a number).

WORKED EXAMPLE T1 Q11 · Odd square iff odd integer · Proposition 4.6.4

For $n \in \mathbb{Z}$, a proposed “contradiction proof” shows only that n even implies n^2 even. **Faults:** it uses contraposition, proves only one direction, and omits justifications. A complete proof is:

1. ($\Rightarrow$) Suppose n is even: $n = 2k$, $k \in \mathbb{Z}$. Then $n^2 = 2(2k^2)$ is even (*definition; closure*). By contraposition, n^2 odd implies n odd.
2. ($\Leftarrow$) If $n = 2k + 1$, $k \in \mathbb{Z}$, then $n^2 = 2(2k^2 + 2k) + 1$ is odd (*algebra; closure*). Hence n^2 is odd iff n is odd. ■

Proposition 4.6.4: n^2 even implies n even, by contraposition of line 2 and the even/odd dichotomy.

10 · Definitions and results in number theory

DEFINITION Even, odd · Prime, composite

$n \in \mathbb{Z}$ is **even** $\Leftrightarrow \exists k \in \mathbb{Z} (n = 2k)$; **odd** $\Leftrightarrow \exists k \in \mathbb{Z} (n = 2k + 1)$.

n is **prime** iff $n > 1$ and $\forall r, s \in \mathbb{Z}^+ (n = rs \rightarrow (r = 1 \wedge s = n) \vee (r = n \wedge s = 1))$.

n is **composite** iff $n > 1$ and $\exists r, s \in \mathbb{Z}^+ (n = rs \wedge 1 < r < n \wedge 1 < s < n)$. 1 is neither.

An equivalent definition of prime: $n > 1 \wedge \forall r, s \in \mathbb{Z} (r > 1 \wedge s > 1 \rightarrow rs \neq n)$.

DEFINITION Rational, irrational · Lowest terms

$r \in \mathbb{R}$ is **rational** $\Leftrightarrow \exists a, b \in \mathbb{Z} (r = a/b \wedge b \neq 0)$; otherwise **irrational**. A fraction a/b is in **lowest terms** if the largest integer dividing both a and b is 1.

DEFINITION Divisibility

For $n, d \in \mathbb{Z}$: $d \mid n \Leftrightarrow \exists k \in \mathbb{Z} (n = dk)$. Read: d divides n ; n is a multiple of d ; d is a factor/divisor of n .

Consequences: every d divides 0; 0 divides only 0; $1 \mid n$ and $n \mid n$ for all n . No division is performed: $3 \mid 12$ is **true** and $3 \mid 10$ is **false**.

DEFINITION Absolute value (Lecture 1) · Colorful (Lecture 1, CS1231S only)

For $x \in \mathbb{R}$, the **absolute value** of x is $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

An integer n is **colorful** iff there exists some integer k such that $n = 3k$. (Non-standard term, used only in the lecture: $-1353 = 3 \cdot (-451)$ and $0 = 3 \cdot 0$ are colorful; 7 is not.)

THEOREM Theorem 4.4.1 · Quotient–Remainder · div and mod

For $n \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$ there exist **unique** $q, r \in \mathbb{Z}$ with $n = dq + r$ and $0 \leq r < d$. We write $n \text{ div } d = q$ and $n \text{ mod } d = r$. Hence every integer is of exactly one form $dq, dq + 1, \dots, dq + (d - 1)$. This is the basis for **division into cases**.

Slide-numbered results and proofs: §12; theorem locator: §34; Appendix A: §33.

11 · Choosing the method

To show	Method	Skeleton
$\exists x \in D P(x)$	Construction	Name a witness $x = \dots$; verify $x \in D$ and $P(x)$. There is no need to explain how it was found.
$\sim \forall x \in D (P(x) \rightarrow Q(x))$	Counterexample	One x with $P(x)$ true and $Q(x)$ false. One suffices.
$\forall x \in D P(x)$, D finite	Exhaustion	Check every case explicitly.
$\forall x \in D (P(x) \rightarrow Q(x))$	Direct (generic particular)	Let $x \in D$ be arbitrary with $P(x)$. Unpack definitions; rebuild $Q(x)$.
same	Division into cases	Split by parity / by $x \text{ mod } d$ (QR theorem) / by sign; prove each case.
same	Contraposition	Prove $\forall x (\sim Q(x) \rightarrow \sim P(x))$ directly.
any statement S	Contradiction	Suppose $\sim S$. Derive a contradiction. Conclude S .
$P \Leftrightarrow Q$	Two directions	Prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately.
$\exists! x P(x)$	Existence + uniqueness	Give x with $P(x)$; then if $P(y)$, show $y = x$.

METHOD Contradiction or contraposition?

Use **contraposition** when the negated conclusion is easier to work with; “ n is odd” is easier to use than “ n^2 is not odd”. Use **contradiction** for statements such as “there is no ...”, “there are infinitely many ...” and “... is irrational”, whose negations assert that some object exists.

WORKED EXAMPLE T2 Q8 · Cases inside a conditional

Prove $\forall x \in \mathbb{R} ((x^2 > x) \rightarrow ((x < 0) \vee (x > 1)))$.

- Let $r \in \mathbb{R}$ with $r^2 > r$. Then $r(r - 1) > 0$ (algebra).
- By Appendix A, Theorem T25, r and $r - 1$ have the same sign. If both are positive, $r - 1 > 0$, so $r > 1$. If both are negative, $r < 0$.
- Thus $r < 0$ or $r > 1$; the statement follows since r was arbitrary. ■

The case $x^2 \leq x$ needs no proof: the implication is vacuously true. A graph alone is not a proof.

PAST PAPER T2 Q11 · A factor is at most $\sqrt{n}$

For $n, a, b \in \mathbb{Z}^+$, prove $n = ab \rightarrow (a \leq \sqrt{n} \vee b \leq \sqrt{n})$. **Contraposition:** suppose $a > \sqrt{n}$ and $b > \sqrt{n}$. Then $ab > \sqrt{n} \cdot \sqrt{n} = n$ (Appendix A, Theorem T27), hence $n \neq ab$. ■

Contradiction: suppose $n = ab$ and both factors exceed $\sqrt{n}$. The same inequality gives $ab > n$, contradicting $ab = n$. ■

TRAP T2 Q4 · Closure claims

$\mathbb{Z}$ is **not** closed under division ($3/2 = 1.5 \notin \mathbb{Z}$). Division is not a binary operation on **all** of $\mathbb{Q} \times \mathbb{Q}$, since $r/0$ is undefined; division of a rational by a **nonzero** rational does give a rational. In proving $\mathbb{Q}$ closed under $+$, the line “ $bd \neq 0$ (zero product property)” is required, and is commonly omitted.

12 · Model proofs

PROOF Sum of two even integers is even (Lecture 4, Ex. 4)

1. Let m and n be arbitrary even integers.
 - 1.1. Then $m = 2r$ and $n = 2s$ for some integers r and s (by definition of even).
 - 1.2. $m + n = 2r + 2s = 2(r + s)$ (by basic algebra).
 - 1.3. $r + s$ is an integer (by closure of integers under $+$), so $m + n$ is even (by definition of even).
2. Therefore the sum of any two even integers is even. ■

Lecture 3.4.3: “by definition” and “by closure” apply universal modus ponens: instantiate the general rule at the chosen integers, then use its hypothesis.

WORKED EXAMPLE Odd products and contraposition · Lecture 1, Ex. 1

Prove that the product of two odd integers is odd; deduce that an even product has an even factor.

1. Let $a = 2r + 1$ and $b = 2s + 1$, where $r, s \in \mathbb{Z}$ (definition of odd).
2. $ab = 2(2rs + r + s) + 1$, with $2rs + r + s \in \mathbb{Z}$ (algebra; closure). Thus ab is odd.
3. By contraposition and the even/odd dichotomy, ab even implies a even or b even. ■

In particular, two consecutive odd integers have odd product. Taking $s = r + 1$ gives $ab = 2(2r^2 + 4r + 1) + 1$.

PROOF Theorem 4.2.2 · The sum of any two rational numbers is rational

1. Let r and s be arbitrary rational numbers.
 - 1.1. Then $r = a/b$ and $s = c/d$ for some integers a, b, c, d with $b \neq 0, d \neq 0$ (by definition of rational).
 - 1.2. $r + s = (ad + bc)/(bd)$ (by basic algebra).
 - 1.3. $ad + bc$ and bd are integers (by closure under $+$ and $\times$), and $bd \neq 0$ (zero product property), so $r + s$ is rational (by definition of rational).
2. Therefore the sum of any two rational numbers is rational. ■

Corollary 4.2.3: if $r \in \mathbb{Q}$, then $2r = r + r \in \mathbb{Q}$ by this theorem.

PROOF Theorem 4.3.1 · For all $a, b \in \mathbb{Z}^+$, if $a \mid b$ then $a \leq b$

1. Let a, b be positive integers with $a \mid b$.
 - 1.1. Then $b = ak$ for some integer k (by definition of divisibility).
 - 1.2. Since a and b are positive, k is positive, i.e. $k \geq 1$.
 - 1.3. Therefore $a \leq ak = b$. ■

PROOF Theorem 4.3.2 · The only divisors of 1 are 1 and -1 (division into cases)

1. Let m be any integer with $m \mid 1$. Then $1 = mk$ for some integer k (by definition of divisibility).
 - 1.1. Since $mk > 0$, m and k are both positive or both negative.
 - 1.2. **Case 1:** both positive. Then $m \leq 1$ (by Theorem 4.3.1), so $m = 1$.
 - 1.3. **Case 2:** both negative. Then $1 = (-m)(-k)$, so $-m$ is a positive divisor of 1; as in Case 1, $-m = 1$, i.e. $m = -1$.
2. Therefore the only divisors of 1 are 1 and -1 . ■

PROOF Theorem 4.3.3 · Transitivity of divisibility

1. Suppose a, b, c are integers with $a \mid b$ and $b \mid c$.
 - 1.1. Then $b = ar$ and $c = bs$ for some integers r, s (by definition of divisibility).
 - 1.2. $c = bs = (ar)s = a(rs)$ (by substitution and associativity).
 - 1.3. rs is an integer (by closure under $\times$), so $a \mid c$ (by definition of divisibility). ■

PROOF Every integer n has $n^2 \bmod 4 \in \{0, 1\}$ (division into cases, via parity)

1. Let n be any integer. n is even or odd (*Assumption 1*).
 - 1.1. **Case 1:** $n = 2k$ for some integer k . Then $n^2 = 4k^2 = 4(k^2) + 0$, and $k^2 \in \mathbb{Z}$ (*closure*).
 - 1.2. **Case 2:** $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$, and $k^2 + k \in \mathbb{Z}$ (*closure*).
2. In either case $n^2 = 4q + r$ with $r \in \{0, 1\}$, so $n^2 \bmod 4 \in \{0, 1\}$ (*by the quotient-remainder theorem, r is unique*). ■

PROOF Theorem 4.6.1 · There is no greatest integer (contradiction)

1. Suppose not; that is, there is a greatest integer g , so $g \geq n$ for every integer n .
 - 1.1. Let $G = g + 1$. Then G is an integer (*by closure under $+$*) and $G > g$.
 - 1.2. This contradicts that g is the greatest integer.
2. Hence the supposition is false, and there is no greatest integer. ■

PROOF Theorem 4.7.1 · $\sqrt{2}$ is irrational (contradiction)

1. Suppose not; that is, $\sqrt{2}$ is rational.
 - 1.1. Then $\sqrt{2} = a/b$ for some integers a, b with $b \neq 0$ (*by definition of rational*).
 - 1.2. Reduce a/b to lowest terms m/n (*Assumption 2*) then $m^2 = 2n^2$ (*by basic algebra*).
 - 1.3. So m^2 is even (*by definition of even, as $n^2 \in \mathbb{Z}$*), hence m is even (*by Proposition 4.6.4*).
 - 1.4. Let $m = 2k$. Then $4k^2 = 2n^2$, so $n^2 = 2k^2$ is even, hence n is even (*by Proposition 4.6.4*).
 - 1.5. So 2 divides both m and n , contradicting that m/n is in lowest terms.
2. Therefore $\sqrt{2}$ is irrational. ■

PROOF rational + irrational is irrational (contradiction)

1. Suppose not: some rational r and irrational s have $r + s$ rational.
 - 1.1. Then $r = a/b$ and $r + s = c/d$ with $a, b, c, d \in \mathbb{Z}$, $b, d \neq 0$ (*by definition of rational*).
 - 1.2. So $s = c/d - a/b = (bc - ad)/(bd)$, with $bc - ad, bd \in \mathbb{Z}$ and $bd \neq 0$ (*closure; zero product property*).
 - 1.3. Hence s is rational (*by definition of rational*), contradicting that s is irrational. ■

PROOF The set of primes is infinite (contradiction)

1. Suppose not: the primes are exactly $p_1, p_2, \dots, p_n$. Let $N = p_1 p_2 \cdots p_n + 1$.
 - 1.1. $N > 1$, so some prime p divides N (*existence of a prime divisor, proved below*) $p = p_i$ for some i (*the list is complete*).
 - 1.2. p_i divides $p_1 \cdots p_n = N - 1$ as well. But a prime dividing a cannot divide $a + 1$ (*consecutive-integer divisibility result, proved below*). Contradiction.
2. Hence there are infinitely many primes. ■

PROOF There exist irrational p, q with p^q rational (Lecture 1, Ex. 7; non-constructive proof by cases)

1. $\sqrt{2}$ is irrational (*Theorem 4.7.1*). Consider $\sqrt{2}^{\sqrt{2}}$: it is rational or irrational.
 - 1.1. **Case 1:** rational. Take $p = q = \sqrt{2}$.
 - 1.2. **Case 2:** irrational. Take $p = \sqrt{2}^{\sqrt{2}}$, $q = \sqrt{2}$: then $p^q = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, rational.
2. In either case the required p and q exist. ■

PROOF Any two consecutive integers have opposite parity

1. Let m and $m + 1$ be two arbitrary consecutive integers. m is even or odd (*Assumption 1*).
 - 1.1. **Case 1:** $m = 2k$ for some $k \in \mathbb{Z}$. Then $m + 1 = 2k + 1$ is odd (*by definition of odd*).
 - 1.2. **Case 2:** $m = 2k + 1$. Then $m + 1 = 2(k + 1)$ with $k + 1 \in \mathbb{Z}$ (*closure*), so $m + 1$ is even.
2. In both cases m and $m + 1$ have opposite parity. ■

PROOF The square of any odd integer has the form $8m + 1$ (division into cases via QR)

1. Let n be a arbitrary odd integer. By the QR theorem with $d = 4$, $n = 4q, 4q + 1, 4q + 2$ or $4q + 3$ for some $q \in \mathbb{Z}$; as n is odd, $n = 4q + 1$ or $n = 4q + 3$.
 - 1.1. **Case 1:** $n^2 = 16q^2 + 8q + 1 = 8(2q^2 + q) + 1$.
 - 1.2. **Case 2:** $n^2 = 16q^2 + 24q + 9 = 8(2q^2 + 3q + 1) + 1$.
2. In both cases $n^2 = 8m + 1$ for an integer m (*closure*). ■

PROOF Theorem 4.4.6 · Triangle inequality $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$

1. By Lemma 4.4.4, $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$; adding, $-(|x| + |y|) \leq x + y \leq |x| + |y|$.
 - 1.1. **Case 1:** $x + y \geq 0$. Then $|x + y| = x + y \leq |x| + |y|$ (*definition of absolute value*).
 - 1.2. **Case 2:** $x + y < 0$. Then $|x + y| = -(x + y) \leq |x| + |y|$ (*from the left inequality*).
2. So $|x + y| \leq |x| + |y|$ in all cases. ■

PROOF For any integer a and prime p : if $p \mid a$ then $p \nmid (a+1)$ (contradiction)

1. Suppose not: $p \mid a$ and $p \mid (a+1)$ for some integer a and prime p .
 - 1.1. Then $a = pr$ and $a+1 = ps$ for some $r, s \in \mathbb{Z}$ (definition of divisibility), so $1 = ps - pr = p(s-r)$.
 - 1.2. So $p \mid 1$ (as $s-r \in \mathbb{Z}$), hence $p = 1$ or $p = -1$ (Theorem 4.3.2). This is a contradiction, since a prime is greater than 1.

PROOF If $a \mid b$ and $a \mid c$ then $a \mid (mb + nc)$ for all integers m, n

1. Suppose $a \mid b$ and $a \mid c$: $b = ar$ and $c = as$ for some $r, s \in \mathbb{Z}$ (definition of divisibility).
2. Then $mb + nc = a(mr + ns)$ (basic algebra), and $mr + ns \in \mathbb{Z}$ (closure), so $a \mid (mb + nc)$.

PROOF Lecture 1 · If x and y are “colorful” (n colorful $\leftrightarrow \exists k \in \mathbb{Z} (n = 3k)$), so is $x + 2y$

Work backwards first: the goal “ $x + 2y$ is colorful” means we need an integer c with $x + 2y = 3c$.

1. Let x, y be colorful integers. Then $x = 3a$ and $y = 3b$ for some $a, b \in \mathbb{Z}$ (by definition of colorful).
2. $x + 2y = 3a + 6b = 3(a + 2b)$ (basic algebra). Let $c = a + 2b$, an integer (closure).
3. So $x + 2y = 3c$, and $x + 2y$ is colorful (by definition of colorful).

WORKED EXAMPLE Disproof by counterexample

- “ $\forall a, b, c \in \mathbb{Z} (a \mid bc \rightarrow a \mid b \vee a \mid c)$ ” is false: $4 \mid 2 \cdot 2$ but $4 \nmid 2$.
- “ $\forall a, b \in \mathbb{Z} (a \mid b \wedge b \mid a \rightarrow a = b)$ ” is false: take $a = 1, b = -1$. (True on $\mathbb{Z}^+$: Theorem 4.3.1.)
- “The product of two irrational numbers is irrational” is false: $\sqrt{2} \cdot \sqrt{8} = 4$ (Lecture 1).
- “ $\forall a, b \in \mathbb{R} (a^2 = b^2 \rightarrow a = b)$ ” is false: take $a = 1, b = -1$ (Lecture 4).
- “There is no integer that is both even and odd” is **true**: if $n = 2a = 2b + 1$ then $1 = 2(a - b)$, so $1/2 \in \mathbb{Z}$, a contradiction.

PROOF Theorem 4.2.1 · Every integer is a rational number

1. Let a be an arbitrary integer. Then $a = a/1$, a quotient of integers with non-zero denominator.
2. Hence a is rational (by definition of rational).

PROOF Proof by exhaustion (Lecture 4, Ex. 3) · every even n with $4 \leq n \leq 26$ is a sum of two primes

$4 = 2 + 2, 6 = 3 + 3, 8 = 3 + 5, 10 = 5 + 5, 12 = 5 + 7, 14 = 3 + 11, 16 = 5 + 11, 18 = 7 + 11, 20 = 7 + 13, 22 = 5 + 17, 24 = 5 + 19, 26 = 7 + 19$. Every case is checked.

PROOF For every integer n , $3 \mid (n^3 - n)$ (division into cases, QR with $d = 3$)

1. Let $n \in \mathbb{Z}$. By the QR theorem, $n = 3q, 3q + 1$ or $3q + 2$ for some $q \in \mathbb{Z}$. Note $n^3 - n = (n-1)n(n+1)$ (basic algebra).
 - 1.1. **Case $n = 3q$:** the factor $n = 3q$, so $n^3 - n = 3 \cdot q(n-1)(n+1)$.
 - 1.2. **Case $n = 3q + 1$:** the factor $n - 1 = 3q$, so $n^3 - n = 3 \cdot qn(n+1)$.
 - 1.3. **Case $n = 3q + 2$:** the factor $n + 1 = 3(q + 1)$, so $n^3 - n = 3 \cdot (q + 1)(n - 1)n$.
2. In each case the cofactor is an integer (closure), so $3 \mid (n^3 - n)$ (definition of divisibility).

PROOF Every prime $p > 3$ has the form $6k + 1$ or $6k + 5$

1. Let $p > 3$ be prime. By the QR theorem, $p = 6k + r$ with $r \in \{0, 1, 2, 3, 4, 5\}$.
 - 1.1. If $r \in \{0, 2, 4\}$, then $p = 2(3k + r/2)$ is even and > 2 , so not prime.
 - 1.2. If $r = 3$, then $p = 3(2k + 1)$ with $p > 3$, so not prime.
2. Hence $r = 1$ or $r = 5$.

PROOF Every integer $n > 1$ is divisible by a prime (via well-ordering)

1. Let $n > 1$ be an integer, and let $D = \{d \in \mathbb{Z} : d > 1 \wedge d \mid n\}$. $D \neq \emptyset$, since $n \in D$.
2. D has a smallest element p ($(\mathbb{N}, \leq)$ is well-ordered).
3. p is prime: suppose instead $p = rs$ with $1 < r < p$. Then $r \mid p$ and $p \mid n$, so $r \mid n$ (Theorem 4.3.3), so $r \in D$ with $r < p$, contradicting the minimality of p .
4. Hence the prime p divides n .

PROOF Lemma 4.4.4 · For every real r : $-|r| \leq r \leq |r|$ (division into cases)

1. **Case $r \geq 0$:** $|r| = r$ (definition of absolute value), and $-|r| = -r \leq 0 \leq r$. So $-|r| \leq r = |r|$.
2. **Case $r < 0$:** $|r| = -r > 0$, so $-|r| = r$ and $r < 0 < |r|$. So $-|r| = r \leq |r|$.

PAST PAPER Using unique factorization · AY25 Q6(iii) in full

Unique factorization: every integer $n > 1$ is a product of primes, unique up to order.

Claim: if $n = m^2$ is a perfect square and a prime p divides n , then $p^2 \mid n$.

If $m = 0$, then $n = 0$ and $p^2 \mid 0$. Otherwise write $|m| = p_1^{e_1} \cdots p_k^{e_k}$ (*unique factorization*) then $n = p_1^{2e_1} \cdots p_k^{2e_k}$. A prime dividing n is some p_i (uniqueness of the factorization), and its exponent $2e_i \geq 2$, so $p_i^2 \mid n$. ■

In a perfect square every prime appears to an even power.

WORKED EXAMPLE Division into cases

Prove: for every integer n , $n^2 + n$ is even.

1. Let n be an arbitrary integer. Then n is even or odd (*Assumption 1*).
 - 1.1. **Case 1:** $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, and $2k^2 + k \in \mathbb{Z}$ (*closure*).
 - 1.2. **Case 2:** $n = 2k + 1$. Then $n^2 + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1)$, and $(2k + 1)(k + 1) \in \mathbb{Z}$ (*closure*).
2. In both cases $n^2 + n$ is even (*by definition of even*). ■

WORKED EXAMPLE Contradiction

Prove: there is no smallest positive rational number.

1. Suppose not: let r be the smallest positive rational number.
 - 1.1. $r = a/b$ for some integers a, b with $b \neq 0$ (*definition of rational*). Then $r/2 = a/(2b)$, where $a, 2b \in \mathbb{Z}$ and $2b \neq 0$, so $r/2$ is rational.
 - 1.2. $r/2 > 0$ and $r/2 < r$ (*as $r > 0$*), contradicting the choice of r as the smallest.
2. Hence there is no smallest positive rational number. ■

Part IV · Set Theory

13 · Describing sets

DEFINITION Set · Membership · Cardinality

A **set** is an unordered collection of objects, its **members** or **elements**; order and repetition do not matter: $\{9, 8, 7\} = \{7, 8, 7, 9, 9\}$. $x \in S$ means x is an element of S ; $x \notin S$ means it is not. The **cardinality** $|S|$ is the size of S , the number of elements of S .

DEFINITION Set-roster, set-builder and replacement notation

Set-roster: list all the elements between braces, e.g. $\{1, 2, 3\}$, $\{1, 2, \dots, 100\}$, $\{1, 2, 3, \dots\}$ (the ellipsis “...” reads “and so forth”).

Set-builder: $\{x \in U : P(x)\}$ is the set of all $x \in U$ for which $P(x)$ is true. z is a member iff $z \in U$ and $P(z)$.

Replacement: $\{t(x) : x \in A\}$ is the set of all objects $t(x)$ as x ranges over A . z is a member iff $t(x) = z$ for **some** $x \in A$.

E.g. $\{x + 1 : x \in \mathbb{Z}_{\geq 0}\} = \mathbb{Z}^+$; $\{x \in \mathbb{Z} : -2 < x < 5\} = \{-1, 0, 1, 2, 3, 4\}$.

DEFINITION Interval notation (Lecture 5)

For real numbers $a \leq b$: $(a, b) = \{x \in \mathbb{R} : a < x < b\}$, $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$, $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$, $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$.

Unbounded: $(a, \infty) = \{x \in \mathbb{R} : x > a\}$, $[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$, $(-\infty, b) = \{x \in \mathbb{R} : x < b\}$, $(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$.

The interval (a, b) and the ordered pair (a, b) look the same; context tells them apart.

Symbol	Set
$\mathbb{N}$	natural numbers $\{0, 1, 2, \dots\}$, including 0
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$	integers, rationals, reals
$\mathbb{Z}^+, \mathbb{Z}^-, \mathbb{Z}_{\geq 0}$	positive, negative, non-negative integers (0 is neither positive nor negative)
$(a, b), [a, b]$	open / closed real intervals; also $(a, b], [a, \infty) \dots$

PAST PAPER AY25 Q17(b) · Name these subsets of $\mathbb{N}$

$A = \{0, 1, 3, 7, 15, 31, \dots\}$ in replacement notation:

$A = \{2^n - 1 : n \in \mathbb{N}\}$.

$B = \{3, 9, 15, 21, 27, \dots\}$ in set-builder notation: $B = \{n \in \mathbb{N} : 3 \mid n \wedge n \text{ is odd}\}$.

(Many correct answers; check the first few terms.)

14 · Subsets, equality and the empty set

DEFINITION Subset, superset · Proper subset · Set equality · Empty set, singleton

$A \subseteq B \Leftrightarrow \forall x(x \in A \rightarrow x \in B)$: every element of A is in B (“ A is contained in B ”). $A \not\subseteq B \Leftrightarrow \exists x(x \in A \wedge x \notin B)$.

If $A \subseteq B$ we may write $B \supseteq A$: “ B contains / includes A ”, “ B is a **superset** of A ”.

$A \subsetneq B$ (proper, or strict, inclusion) iff $A \subseteq B$ and $A \neq B$. $A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A$

$\Leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$.

The **empty set** $\emptyset = \{\}$ is the set with no elements. A set with exactly one element is a **singleton**.

THEOREM Theorem 6.2.4 · The empty set

$\emptyset \subseteq A$ for every set A (and $\emptyset$ is unique). **Proof:** $\forall x(x \in \emptyset \rightarrow x \in A)$ is vacuously true, as $x \in \emptyset$ is always false.

Uniqueness: the empty set is unique: two empty sets are subsets of one another by Theorem 6.2.4, hence equal.

Claim		Why
$\emptyset \in \{1, 2, 3\}$	F	$\emptyset$ is not listed
$\emptyset \subseteq \{1, 2, 3\}$	T	Thm 6.2.4
$\{2\} \in \{1, 2, 3\}$	F	elements are numbers
$\{2\} \subseteq \{1, 2, 3\}$	T	2 is an element
$\{2\} \in \{\{1\}, \{2\}, \{3\}\}$	T	$\{2\}$ is listed
$\{2\} \subseteq \{\{1\}, \{2\}, \{3\}\}$	F	2 is not listed
$\{9\} = \{\{9\}\}$	F	different elements

The $\emptyset$ ladder	$ \cdot $
$\emptyset = \{\}$	0
$\{\emptyset\} = \mathcal{P}(\emptyset)$	1
$\{\emptyset, \{\emptyset\}\} = \mathcal{P}(\mathcal{P}(\emptyset))$	2
$\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$	4
$\mathcal{P}^4(\emptyset)$	16

$\emptyset \neq \{\emptyset\}$: the first is empty, the second has one element. $\emptyset \in \{\emptyset\}$ and $\emptyset \subseteq \{\emptyset\}$ are both true.

TRAP $\subseteq$ is transitive but $\in$ is not · AY23 Q8

Which hold for all sets? (i) $A \in B \wedge B \subseteq C \Rightarrow A \in C$ is **true**, since A is an element of B and hence of C .

(ii) $A \in B \wedge B \subseteq C \Rightarrow A \subseteq C$ is false: take $A = \{1\}$, $B = C = \{\{1\}\}$, but $1 \notin C$.

(iii), (iv) $A \subseteq B \wedge B \in C \Rightarrow A \in C$ or $A \subseteq C$ are false: take $A = \{x\}$, $B = \{x, y\}$, $C = \{\{x, y\}\}$. Only (i): **A**.

15 · Ordered pairs and Cartesian products

DEFINITION Ordered pair, n -tuple · Cartesian product

$(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$; likewise for n -tuples componentwise. $(1, 2) \neq (2, 1)$ though $\{1, 2\} = \{2, 1\}$.

$A \times B = \{(a, b) : a \in A \wedge b \in B\}$; $A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i\}$; $A^n = A \times \dots \times A$.

Facts. $|A \times B| = |A| \cdot |B|$. $A \times \emptyset = \emptyset \times A = \emptyset$. $A \times B \neq B \times A$ in general. $(A \times B) \times C \neq A \times B \times C$: elements $((a, b), c)$ versus (a, b, c) . $A \times (B \cup C) = (A \times B) \cup (A \times C)$, and likewise for $\cap$ and $\setminus$.

PAST PAPER AY24 Q6 · Which hold for all sets A, B, C ?

(i) $\mathcal{P}(A) \subseteq \mathcal{P}(B) \rightarrow A \subseteq B$ is **true** (proof in §17). (ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ is **true** (proof in §17).

(iii) $A \times B \subseteq B \times C \rightarrow A \subseteq C$ is **false**: take $B = \emptyset$; then $A \times \emptyset = \emptyset$ is a subset of every set, whatever A is.

Answer **A**.

TRAP T3 Q3(c) · $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$

For every A, B , $\emptyset \in \mathcal{P}(A \times B)$ ($\emptyset \subseteq A \times B$), but $\emptyset \notin A \times \mathcal{P}(B)$ (every member of a Cartesian product is an ordered pair; $\emptyset$ is not one). This named witness proves the sets differ. A cardinality “proof” ($n \cdot 2^k \neq 2^{nk}$ for $|A| = n$, $|B| = k$) is wrong: equality holds when $n = 1$, or $n = 2$ and $k = 1$.

16 · Operations on sets

DEFINITION Universal set · Union, intersection, difference, complement

A **universal set** (universe of discourse) U is the set containing every object under discussion, e.g. $\mathbb{R}$ when all sets considered are sets of reals.

For $A, B \subseteq U$: $A \cup B = \{x \in U : x \in A \vee x \in B\}$, $A \cap B = \{x \in U : x \in A \wedge x \in B\}$,
 $B \setminus A = \{x \in U : x \in B \wedge x \notin A\}$ (the **relative complement** of A in B), $\overline{A} = \{x \in U : x \notin A\}$
 $= U \setminus A$.

Procedural versions (the steps in element proofs): $a \in X \cup Y \Leftrightarrow a \in X \vee a \in Y$; $a \in X \cap Y \Leftrightarrow a \in X \wedge a \in Y$;
 $a \in X \setminus Y \Leftrightarrow a \in X \wedge a \notin Y$; $a \in \overline{X} \Leftrightarrow a \notin X$; $(a, b) \in X \times Y \Leftrightarrow a \in X \wedge b \in Y$.

DEFINITION Symmetric difference (Tutorial 3) · Disjoint

$A \oplus B = (A \setminus B) \cup (B \setminus A)$, the set of elements in exactly one of A and B . A and B are **disjoint** iff $A \cap B = \emptyset$; $A_1, A_2, \dots$ are **mutually disjoint** iff $A_i \cap A_j = \emptyset$ whenever $i \neq j$. $\bigcup_{i=0}^n A_i, \bigcap_{i=0}^n A_i$ extend $\cup, \cap$.

THEOREM Theorem 6.2.1 · Subset relations

$A \cap B \subseteq A, A \cap B \subseteq B$ (inclusion of intersection); $A \subseteq A \cup B, B \subseteq A \cup B$ (inclusion in union);

$A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$ (transitivity). The **set identities** (Theorem 6.2.2) are the right column of the table in §2.

PAST PAPER AY23 Q7 · For which can no non-empty A, B exist?

- (i) $A \setminus B = A \cap B$: these two sets are disjoint, so both must be $\emptyset$; then $A = (A \setminus B) \cup (A \cap B) = \emptyset$. **Impossible.**
- (ii) $A \setminus B = A \cup B$: then $B \subseteq A \cup B = A \setminus B$, so every element of B is not in B ; $B = \emptyset$. **Impossible.**
- (iii) $A \cup B = A \cap B$: take $A = B$. (iv) $A \oplus B = A \cup B$: take $A \cap B = \emptyset$. Answer **B** (only (i), (ii)).

17 · Proving and disproving set statements

METHOD Three methods

Element method. To show $X \subseteq Y$: “Let $z \in X$... so $z \in Y$ ”. To show $X = Y$: prove both inclusions, or one chain of $\Leftrightarrow$ steps each justified by a definition or a logic law.

Algebraic method. Rewrite using the set identities (§2), one named law per line.

Disproof. Give one concrete counterexample with every set written out. A Venn diagram helps in finding one.

PROOF $\{x \in \mathbb{Z} : x^2 = 1\} = \{1, -1\}$ (two inclusions, Lecture 5)

1. ($\subseteq$) Take any $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
 - 1.1. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $(z-1)(z+1) = 0$ (by basic algebra).
 - 1.2. So $z = 1$ or $z = -1$, i.e. $z \in \{1, -1\}$.
2. ($\supseteq$) Take any $z \in \{1, -1\}$. Then $z \in \mathbb{Z}$ and $z^2 = 1$, so $z \in \{x \in \mathbb{Z} : x^2 = 1\}$.
3. Therefore the sets are equal (by definition of set equality, from 1 and 2). ■

PROOF De Morgan: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (chain of $\Leftrightarrow$)

1. Let $z \in U$. Then
 - 1.1. $z \in \overline{A \cup B} \Leftrightarrow \sim(z \in A \cup B)$ (by definition of complement)
 - 1.2. $\Leftrightarrow \sim(z \in A \vee z \in B)$ (by definition of $\cup$)
 - 1.3. $\Leftrightarrow z \notin A \wedge z \notin B$ (by De Morgan's law for logic)
 - 1.4. $\Leftrightarrow z \in \overline{A} \wedge z \in \overline{B} \Leftrightarrow z \in \overline{A} \cap \overline{B}$ (by definitions of complement, $\cap$). ■

PROOF $(A \cap B) \cup (A \setminus B) = A$ (algebraic, Lecture 5)

$$\begin{aligned}
 (A \cap B) \cup (A \setminus B) &= (A \cap B) \cup (A \cap \overline{B}) && \text{(set difference law)} \\
 &= A \cap (B \cup \overline{B}) && \text{(distributive law)} \\
 &= A \cap U = A && \text{(complement law; identity law)}
 \end{aligned}$$

■

PROOF $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$ (algebraic)

$$\begin{aligned}
 A \setminus (B \cap C) &= A \cap \overline{B \cap C} && \text{(set difference)} \\
 &= A \cap (\overline{B} \cup \overline{C}) && \text{(De Morgan)}
 \end{aligned}$$

$$\begin{aligned}
&= (A \cap \overline{B}) \cup (A \cap \overline{C}) \quad (\text{distributive}) \\
&= (A \setminus B) \cup (A \setminus C) \quad (\text{set difference } \times 2).
\end{aligned}$$

PROOF For all sets A, B : $A \subseteq B \Leftrightarrow \mathcal{P}(A) \subseteq \mathcal{P}(B)$

1. $(\Rightarrow)$ Suppose $A \subseteq B$. Let $X \in \mathcal{P}(A)$; then $X \subseteq A$ (definition of power set), so $X \subseteq B$ (transitivity of $\subseteq$), so $X \in \mathcal{P}(B)$.
2. $(\Leftarrow)$ Suppose $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. $A \subseteq A$, so $A \in \mathcal{P}(A)$, hence $A \in \mathcal{P}(B)$, i.e. $A \subseteq B$ (definition of power set).

WORKED EXAMPLE T3 Q4 · Set equality by double inclusion

Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$. Prove $A = B$.

1. If $a \in A$, write $a = 2n + 1 = 2(n + 3) - 5$ for $n \in \mathbb{Z}$. Since $n + 3 \in \mathbb{Z}$ (closure), $a \in B$ (definition of B).
2. If $b \in B$, write $b = 2n - 5 = 2(n - 3) + 1$ for $n \in \mathbb{Z}$. Since $n - 3 \in \mathbb{Z}$, $b \in A$.
3. Thus $A = B$ (double inclusion).

A substitution $k = n + 3$ must range over **all** integers: given k , use $n = k - 3$. Closure alone does not show this.

WORKED EXAMPLE T3 Q5 · Element method as a set-builder chain

Prove that for all sets A, B, C : $A \cap (B \setminus C) = (A \cap B) \setminus C$.

$$\begin{aligned}
A \cap (B \setminus C) &= \{x : x \in A \wedge x \in (B \setminus C)\} \quad (\text{definition of } \cap) \\
&= \{x : x \in A \wedge (x \in B \wedge x \notin C)\} \quad (\text{definition of } \setminus) \\
&= \{x : (x \in A \wedge x \in B) \wedge x \notin C\} \quad (\text{associativity of } \wedge) \\
&= \{x : x \in A \cap B \wedge x \notin C\} \quad (\text{definition of } \cap) \\
&= (A \cap B) \setminus C \quad (\text{definition of } \setminus).
\end{aligned}$$

WORKED EXAMPLE T3 Q6 · Symmetric difference by set identities only

(a) $A = \{1, 4, 9, 16\}$, $B = \{2, 4, 6, \dots, 16\}$: find $A \oplus B$. (b) Using Theorem 6.2.2, prove $A \oplus B = (A \cup B) \setminus (A \cap B)$.

(a) $A \setminus B = \{1, 9\}$, $B \setminus A = \{2, 6, 8, 10, 12, 14\}$, so $A \oplus B = \{1, 2, 6, 8, 9, 10, 12, 14\}$.

$$\begin{aligned}
\text{(b) } A \oplus B &= (A \setminus B) \cup (B \setminus A) \quad (\text{definition of } \oplus) \\
&= (A \cap \overline{B}) \cup (B \cap \overline{A}) \quad (\text{set difference law}) \\
&= ((A \cap \overline{B}) \cup B) \cap ((A \cap \overline{B}) \cup \overline{A}) \quad (\text{distributive law}) \\
&= ((A \cup B) \cap (\overline{B} \cup B)) \cap ((A \cup \overline{A}) \cap (\overline{B} \cup \overline{A})) \quad (\text{distributive, commutative laws}) \\
&= ((A \cup B) \cap U) \cap (U \cap (\overline{A} \cup \overline{B})) \quad (\text{commutative, complement laws}) \\
&= (A \cup B) \cap (\overline{A} \cup \overline{B}) \quad (\text{commutative, identity laws}) \\
&= (A \cup B) \cap \overline{A \cap B} \quad (\text{De Morgan's law}) \\
&= (A \cup B) \setminus (A \cap B) \quad (\text{set difference law}).
\end{aligned}$$

WORKED EXAMPLE Set identity

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

$$\begin{aligned}
A \setminus (B \cup C) &= A \cap \overline{B \cup C} \quad (\text{set difference law}) \\
&= A \cap (\overline{B} \cap \overline{C}) \quad (\text{De Morgan's law}) \\
&= (A \cap A) \cap (\overline{B} \cap \overline{C}) \quad (\text{idempotent law}) \\
&= (A \cap \overline{B}) \cap (A \cap \overline{C}) \quad (\text{associative, commutative laws}) \\
&= (A \setminus B) \cap (A \setminus C) \quad (\text{set difference law}).
\end{aligned}$$

WORKED EXAMPLE Prove or disprove

(a) $A \cup (B \setminus C) = (A \cup B) \setminus C$; (b) $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$; (c) $A \subseteq C \wedge B \subseteq C \rightarrow A \cup B \subseteq C$.

(a) **False:** $A = \{1\}$, $B = \emptyset$, $C = \{1\}$: left side $\{1\}$, right side $\emptyset$.

(b) **False:** $A = \{1\}$, $B = \{2\}$: $\{1, 2\} \in \mathcal{P}(A \cup B)$ but $\{1, 2\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$. (Only $\supseteq$ holds.)

(c) **True:** let $x \in A \cup B$; then $x \in A$ or $x \in B$ (definition of $\cup$) in either case $x \in C$ (as $A \subseteq C$, $B \subseteq C$).

18 · Partitions and power sets

DEFINITION Partition (Lecture 6)

$\mathcal{C}$ is a **partition** of A iff (1) every element of $\mathcal{C}$ is a **non-empty** subset of A , and (2) every element of A lies in **exactly one** element of $\mathcal{C}$: $\forall x \in A \exists! S \in \mathcal{C} (x \in S)$. The elements of $\mathcal{C}$ are its **components**. (Equivalently: non-empty, mutually disjoint, union is A .)

DEFINITION Power set · Theorem 6.3.1

$\mathcal{P}(A)$ is the set of all subsets of A . If A is finite and $|A| = n$, then $|\mathcal{P}(A)| = 2^n$. Always $\emptyset \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.

$\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, but only $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

PAST PAPER AY24 Q7 and Q8 · Partitions, power sets

Q7. (i) $\{\{2, 3\}, \{4\}, \{3, 2\}, \{1\}\} = \{\{2, 3\}, \{4\}, \{1\}\}$ is a partition of $\{1, 2, 3, 4\}$, because repeated elements count once. (ii) $\{\mathbb{Z}\}$ is a partition of $\mathbb{Z}$ (one component). (iii) $\mathbb{Z}$ is **not**: its elements are numbers, not subsets. **C.**

Q8. (i) $|\mathcal{P}(S) \setminus \mathcal{P}(\emptyset)| = 2^{|S|} - 1$ is true, as $\mathcal{P}(\emptyset) = \{\emptyset\}$. (ii) $|S \cap \mathcal{P}(S)| = 0$ is false: $S = \{a, \{a\}\}$ gives $S \cap \mathcal{P}(S) = \{\{a\}\}$. (iii) $|S| \cdot |\mathcal{P}(S)| = |S \times \mathcal{P}(S)|$ is true. **D.**

PAST PAPER AY25 Q7, Q8 · Counting

Q7. A, B, C pairwise share exactly one element; $D = \mathcal{P}(A) \cap \mathcal{P}(B) \cap \mathcal{P}(C) = \mathcal{P}(A \cap B \cap C)$. The triple intersection lies inside each one-element pairwise intersection, so it has 0 or 1 elements: $|D| \in \{1, 2\}$. **B.**

Q8. $|A| = 2|B|$, $|B| = |C| + 2$, and $\{B, C, D\}$ partitions A , so $|A| = |B| + |C| + |D|$: $2|B| = |B| + (|B| - 2) + |D|$, giving $|D| = 2$. **C.**

PAST PAPER AY24 Q16 (5 marks) · $A = \{\emptyset, x\}$

$\mathcal{P}(A) = \{\emptyset, \{\emptyset\}, \{x\}, \{\emptyset, x\}\}$ and $\mathcal{P}(\emptyset) = \{\emptyset\}$, $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$.

- (a) $\mathcal{P}(A) \setminus \mathcal{P}(\mathcal{P}(\emptyset)) = \{\{x\}, \{\emptyset, x\}\}$.
- (b) $(\mathcal{P}(A) \setminus \mathcal{P}(\emptyset)) \setminus \{A\} = \{\{\emptyset\}, \{x\}\}$, so its power set is $\{\emptyset, \{\{\emptyset\}\}, \{\{x\}\}, \{\{\emptyset\}, \{x\}\}\}$.
- (c) $\mathcal{P}(\emptyset) \times \mathcal{P}(A) = \{(\emptyset, S) : S \in \mathcal{P}(A)\}$ and $\mathcal{P}(A) \times \mathcal{P}(\emptyset) = \{(S, \emptyset) : S \in \mathcal{P}(A)\}$ share only $(\emptyset, \emptyset)$, so the answer is $\{(\emptyset, \{\emptyset\}), (\emptyset, \{x\}), (\emptyset, \{\emptyset, x\})\}$.

PAST PAPER AY23 Q6 and Q17 · $A_1 = \{\emptyset\}$, $A_2 = \{a\}$, $A_3 = \{a, \{a\}\}$

Q6. Target $\emptyset \cup \mathcal{P}(\emptyset) = \{\emptyset\}$. $\emptyset \times \{\emptyset\} = \emptyset \times \emptyset = \emptyset$; $\mathcal{P}(\emptyset \cup \mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$; $\mathcal{P}(\emptyset) \setminus \emptyset = \{\emptyset\}$ ✓; $(\emptyset \cap \{\emptyset\}) \cup (\emptyset \cup \{\emptyset\}) = \{\emptyset\}$ ✓. **D.**

Q17. (a) $(A_1 \cap A_2) \setminus A_3 = \emptyset$. (b) $(A_1 \cup A_2) \setminus A_3 = \{\emptyset\}$. (c) $(A_1 \times A_2) \cup A_3 = \{(\emptyset, a), a, \{a\}\}$. (d) $(A_1 \cup A_2) \times A_3 = \{(\emptyset, a), (\emptyset, \{a\}), (a, a), (a, \{a\})\}$.

PROOF Distributive law $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, by the element method with cases

1. ($\subseteq$) Let $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$ (definition of $\cup$).
 - 1.1. **Case $x \in A$:** then $x \in A \cup B$ and $x \in A \cup C$ (definition of $\cup$), so $x \in (A \cup B) \cap (A \cup C)$.
 - 1.2. **Case $x \in B \cap C$:** then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$.
2. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$, so $x \in A \cup B$ and $x \in A \cup C$.
 - 2.1. **Case $x \in A$:** then $x \in A \cup (B \cap C)$.
 - 2.2. **Case $x \notin A$:** then $x \in B$ and $x \in C$ (as $x \in A \cup B$, $x \in A \cup C$), so $x \in B \cap C \subseteq A \cup (B \cap C)$.
3. Hence the two sets are equal (definition of set equality). ■

PROOF For all sets A, B : $A \subseteq B \leftrightarrow A \cup B = B$

1. ($\rightarrow$) Suppose $A \subseteq B$. $B \subseteq A \cup B$ (inclusion in union). Conversely let $x \in A \cup B$: if $x \in A$ then $x \in B$ (as $A \subseteq B$) if $x \in B$, done. So $A \cup B = B$.
2. ($\leftarrow$) Suppose $A \cup B = B$. Let $x \in A$; then $x \in A \cup B$ (inclusion in union) $= B$. So $A \subseteq B$. ■

PROOF For every set A , $A \times \emptyset = \emptyset$ (proof by contradiction)

1. Suppose not: some $z \in A \times \emptyset$. Then $z = (a, b)$ with $a \in A$ and $b \in \emptyset$ (definition of $\times$).
2. But $\emptyset$ has no elements, a contradiction. So $A \times \emptyset$ has no elements: $A \times \emptyset = \emptyset$. ■

PROOF $A \times (B \cup C) = (A \times B) \cup (A \times C)$ (chain of $\leftrightarrow$)

$(x, y) \in A \times (B \cup C)$
 $\leftrightarrow x \in A \wedge (y \in B \vee y \in C)$ (definitions of $\times, \cup$)
 $\leftrightarrow (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$ (distributive law of logic)
 $\leftrightarrow (x, y) \in (A \times B) \cup (A \times C)$ (definitions of $\times, \cup$). ■

PROOF $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$ (algebraic)

$(A \cup B) \setminus C$
 $= (A \cup B) \cap \overline{C}$ (set difference law)
 $= (A \cap \overline{C}) \cup (B \cap \overline{C})$ (distributive, commutative)
 $= (A \setminus C) \cup (B \setminus C)$ (set difference law). ■

WORKED EXAMPLE Disproving a set identity · “ $(A \setminus B) \cup (B \setminus C) = A \setminus C$ for all sets”

False. Counterexample: $A = \emptyset$, $B = \{1\}$, $C = \emptyset$. Then $(A \setminus B) \cup (B \setminus C) = \emptyset \cup \{1\} = \{1\}$, while $A \setminus C = \emptyset$. Write out every set and compute both sides.

DRILL Nested sets and power sets (answers after the arrow)

- $|\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}| \rightarrow 3$
- $\mathcal{P}(\{\emptyset\}) \rightarrow \{\emptyset, \{\emptyset\}\}$
- $\{\emptyset\} \times \{\emptyset\} \rightarrow \{(\emptyset, \emptyset)\}$
- $\emptyset \times \{\emptyset\} \rightarrow \emptyset$
- $\mathcal{P}(\{a, b\}) \setminus \{\emptyset\} \rightarrow \{\{a\}, \{b\}, \{a, b\}\}$
- $\{1, \{1\}\} \cap \mathcal{P}(\{1\}) \rightarrow \{\{1\}\}$
- $|\mathcal{P}(A \times B)|$ with $|A| = 2$, $|B| = 3 \rightarrow 2^6 = 64$
- $\{x \in \mathbb{Z} : x^2 < 5\} \rightarrow \{-2, -1, 0, 1, 2\}$
- $\{2n + 1 : n \in \{0, 1, 2\}\} \rightarrow \{1, 3, 5\}$
- $\{\{1, 2\}, \{3\}\}$ a partition of $\{1, 2, 3\}$? $\rightarrow$ **yes**;
 $\{\{1, 2\}, \{2, 3\}\} \rightarrow$ **no** (2 twice); $\{\{1, 2\}, \{3\}, \emptyset\} \rightarrow$
no ($\emptyset$ component)

PROOF If $A \subseteq B$ then $A \cap C \subseteq B \cap C$ and $A \cup C \subseteq B \cup C$

1. Let $x \in A \cap C$: $x \in A$ and $x \in C$ (definition of $\cap$) $x \in B$ (as $A \subseteq B$) so $x \in B \cap C$.
2. Let $x \in A \cup C$: if $x \in A$ then $x \in B \subseteq B \cup C$; if $x \in C$ then $x \in B \cup C$ (inclusion in union). ■

Reading only (Lecture 5.3). Unrestricted comprehension breaks: Russell's $R = \{x : x \notin x\}$ gives $R \in R \leftrightarrow R \notin R$; a set of “everything” $\mathcal{D}$ contradicts $|\mathcal{P}(\mathcal{D})| > |\mathcal{D}|$ (Cantor). Zermelo (1908), later Fraenkel and Skolem, fixed this with the ZF axioms. Cantor, Russell, Zermelo and Fraenkel contributed to modern set theory; Gauss did not (AY25 Q1).

WORKED EXAMPLE Power set of a set that contains $\emptyset$

Let $A = \{\emptyset, \{\emptyset\}\}$. Find $\mathcal{P}(A)$ and $A \cap \mathcal{P}(A)$.

$\mathcal{P}(A) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$, with four subsets since $|A| = 2$. For $A \cap \mathcal{P}(A)$ test each element of A : $\emptyset \in \mathcal{P}(A)$ (as $\emptyset \subseteq A$), and $\{\emptyset\} \in \mathcal{P}(A)$ (as $\{\emptyset\} \subseteq A$, since $\emptyset \in A$). So $A \cap \mathcal{P}(A) = \{\emptyset, \{\emptyset\}\} = A$, i.e. $A \subseteq \mathcal{P}(A)$.

Part V · Relations

19 · Relations and inverses

DEFINITION Relation · Domain, co-domain, range

A (binary) **relation from A to B** is a subset $R \subseteq A \times B$; xRy (“ x is R -related to y ”) means $(x, y) \in R$, and $x \not R y$ means $(x, y) \notin R$. A **relation on A** is a relation from A to A , i.e. a subset of $A \times A = A^2$. Its **directed graph** draws each element of A once, with an arrow $x \rightarrow y$ whenever xRy and a loop at x when xRx .

$$\text{Dom}(R) = \{a \in A : aRb \text{ for some } b \in B\}, \quad \text{coDom}(R) = B, \quad \text{Range}(R) = \{b \in B : aRb \text{ for some } a \in A\}.$$

An **n -ary relation** on $A_1 \times \cdots \times A_n$ is a subset of it (binary, ternary, quaternary for $n = 2, 3, 4$).

Relational databases are built on n -ary relations.

DEFINITION Inverse relation

$R^{-1} = \{(y, x) \in B \times A : (x, y) \in R\}$, i.e. $\forall x \in A, y \in B ((y, x) \in R^{-1} \leftrightarrow (x, y) \in R)$. Reverse every arrow.

PAST PAPER AY25 Q9, Q10 · Relations are sets of pairs

Q9. $A = \{1, 2, 3\}$, $B = \{2, 5\}$. $S_2 = \mathcal{P}(A \times B)$ is **exactly** the set S_3 of relations from A to B ; $S_1 = \mathcal{P}(A) \times \mathcal{P}(B)$ holds **pairs of sets**, a different kind of object. So $S_4 \subseteq S_3 \subseteq S_2$. Answer **A**.

Q10. The relation “ $p \rightarrow q$ ” on $\{T, F\}$ is $R = \{(T, T), (F, T), (F, F)\}$. $R^{-1} = \{(T, T), (T, F), (F, F)\}$ is “ $q \rightarrow p$ ”, the **converse**. The converse is equivalent to the inverse, so the answer key accepted **B or D**.

20 · Composition of relations

DEFINITION Composition (Lecture 6.1.4)

For $R \subseteq A \times B$ and $S \subseteq B \times C$, the **composition** $S \circ R$ is the relation from A to C with

$$\forall x \in A, \forall z \in C (x(S \circ R)z \leftrightarrow \exists y \in B (xRy \wedge ySz)).$$

x and z are related iff there is a path $x \xrightarrow{R} y \xrightarrow{S} z$. **$S \circ R$ applies R first.**

METHOD Computing $S \circ R$ by hand

For each pair $(x, y) \in R$, list every $(y, z) \in S$ that **starts where it ends**, and record (x, z) . For $R \circ R \circ R$, compute $R_2 = R \circ R$ first, then $R \circ R_2$. Stop and re-check pairs that could join through two different middles.

PAST PAPER AY25 Q17(a) (4 marks) · $S = \{(1, 3), (3, 6), (6, 9), (9, 12)\}$, $R = \{(6, 6), (9, 9)\}$

$R \circ S$ (apply S then R): $(3, 6)(6, 6) \rightarrow (3, 6)$; $(6, 9)(9, 9) \rightarrow (6, 9)$. So $R \circ S = \{(3, 6), (6, 9)\}$.

$S \circ R$ (apply R then S): $(6, 6)(6, 9) \rightarrow (6, 9)$; $(9, 9)(9, 12) \rightarrow (9, 12)$. So $S \circ R = \{(6, 9), (9, 12)\}$.

(i) $(R \circ S) \setminus (S \circ R) = \{(3, 6)\}$.

(ii) $(R^{-1} \circ S^{-1}) \circ (S \circ R) = (S \circ R)^{-1} \circ (S \circ R)$; with $(S \circ R)^{-1} = \{(9, 6), (12, 9)\}$ this is $\{(6, 6), (9, 9)\}$.

PAST PAPER AY24 Q11, Q17(e), Q12 · Powers and formula relations

Q11. $R = \{(a, a), (a, b), (b, c), (c, a), (c, b)\}$ on $\{a, b, c\}$. $R \circ R$ is all of $A \times A$ except (b, c) ; then bRc and $c(R \circ R)c$ put (b, c) into $R \circ R \circ R$, which is $A \times A$. Answer **B**.

Q17(e). $xR_1y \leftrightarrow y = 2x$, $xR_2y \leftrightarrow x = y^2$ on $\mathbb{Z}^+$. $x(R_1 \circ R_2)z \leftrightarrow \exists y(x = y^2 \wedge z = 2y)$, so $x = z^2/4$ (with z even).

Q12. $|R \circ R|$ has no fixed relation to $|R|$: on $\{a, b\}$, $R = \{(a, a), (a, b), (b, a)\}$ gives $|R \circ R| = 4 > 3$; $\{(a, a), (a, b)\}$ gives $2 = 2$ but $\neq 2^2$; $\{(a, a)\}$ gives $1 = 1^2$. None of (i)–(iii) always holds. Answer **E**.

PROOF $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ (Lecture 6, slide 18)

Let $R \subseteq A \times B$ and $S \subseteq B \times C$. For all $z \in C$, $x \in A$: $(z, x) \in (S \circ R)^{-1} \leftrightarrow (x, z) \in S \circ R$ (definition of inverse)

$\leftrightarrow \exists y \in B (xRy \wedge ySz)$ (definition of $\circ$)

$\leftrightarrow \exists y \in B (zS^{-1}y \wedge yR^{-1}x)$ (definition of inverse; commutative law)

$\leftrightarrow (z, x) \in R^{-1} \circ S^{-1}$ (definition of $\circ$). ■

PROOF Proposition (Lecture 6, slide 18): $(T \circ S) \circ R = T \circ (S \circ R)$

Let $R \subseteq A \times B$, $S \subseteq B \times C$, $T \subseteq C \times D$, $x \in A$ and $w \in D$.

$x((T \circ S) \circ R)w$

$\leftrightarrow \exists y(xRy \wedge \exists z(ySz \wedge zTw))$

$\leftrightarrow \exists z \exists y(xRy \wedge ySz \wedge zTw)$ (z is not free in xRy ; same-kind quantifiers commute; associative law)

$\leftrightarrow \exists z(\exists y(xRy \wedge ySz) \wedge zTw)$ (y is not free in zTw)

$\leftrightarrow \exists z(x(S \circ R)z \wedge zTw) \leftrightarrow x(T \circ (S \circ R))w$. ■

PAST PAPER T4 Q6 (AY23/24 exam) · Seven copies of an equivalence relation

$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$ on $\{1, 2, 3\}$; find $R \circ R \circ \dots \circ R$ (7 copies). R is an equivalence relation, so $R \circ R = R$ (AY23 Q12, §23). By associativity, $R^7 = ((R \circ R) \circ (R \circ R)) \circ ((R \circ R) \circ R) = (R \circ R) \circ (R \circ R) = R \circ R = R$.

WORKED EXAMPLE T4 Q9 · Inverse and composition by the element method

$S = \{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \text{ is even}\}$. Determine (a) S^{-1} , (b) $S \circ S$, (c) $S \circ S^{-1}$.

(a) $S^{-1} = \{(x, y) \in \mathbb{Z}^2 : (y, x) \in S\}$ (definition of inverse)

$= \{(x, y) \in \mathbb{Z}^2 : y^2 + x^2 \text{ is even}\}$ (definition of S)

$= \{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 \text{ is even}\}$ (commutativity of $+$)

$= S$.

(b) $S \circ S = S$:

1. ($\subseteq$) Let $(x, z) \in S \circ S$. Then $(x, y) \in S$ and $(y, z) \in S$ for some $y \in \mathbb{Z}$ (definition of composition).

1.1. So $x^2 + y^2$ and $y^2 + z^2$ are even (definition of S), hence $x^2 + 2y^2 + z^2$ is even (sum of two even integers).

1.2. As $2y^2$ is even (definition of even, closure), $x^2 + z^2$ is even (difference of two even integers), so $(x, z) \in S$ (definition of S).

2. ($\supseteq$) Let $(x, z) \in S$. Note $(x, x) \in S$, as $x^2 + x^2$ is even. Since $(x, x) \in S$ and $(x, z) \in S$, $(x, z) \in S \circ S$ (definition of composition).

3. Therefore $S \circ S = S$. ■

(c) $S \circ S^{-1} = S \circ S = S$ by (a) and (b).

WORKED EXAMPLE Lecture 6 · Composition as “going through” a middle set

“takes” $\subseteq$ Students $\times$ Modules and “held in” $\subseteq$ Modules $\times$ Venues. Ann takes CS1010, CS1231, MA1101; Bryan takes CS1010, IS1103; Candy takes CS1231, IS1103; Danny takes nothing. CS1010 is held in LT15; CS1231 in ICube and SR1; IS1103 in SR1; CS2100 in LT15; MA1101 nowhere.

“held in $\circ$ takes” = “goes to”: Ann $\rightarrow \{\text{LT15, ICube, SR1}\}$; Bryan $\rightarrow \{\text{LT15, SR1}\}$; Candy $\rightarrow \{\text{ICube, SR1}\}$; Danny $\rightarrow$ nothing. CS2100 contributes no pair, since nobody takes it.

21 · Properties of relations on a set A **DEFINITION** Reflexive, symmetric, transitive (Lecture 6.2) · Antisymmetric (6.4) · Asymmetric (Tutorial 5)

	R has the property iff	R fails it iff
Reflexive	$\forall x \in A (xRx)$	$\exists x \in A (x \not R x)$
Symmetric	$\forall x, y \in A (xRy \rightarrow yRx)$	$\exists x, y (xRy \wedge y \not R x)$
Transitive	$\forall x, y, z \in A (xRy \wedge yRz \rightarrow xRz)$	$\exists x, y, z (xRy \wedge yRz \wedge x \not R z)$
Antisymmetric	$\forall x, y \in A (xRy \wedge yRx \rightarrow x = y)$	$\exists x, y (xRy \wedge yRx \wedge x \neq y)$
Asymmetric	$\forall x, y \in A (xRy \rightarrow y \not R x)$	$\exists x, y (xRy \wedge yRx)$

THEOREM Useful facts

- R symmetric $\Leftrightarrow R = R^{-1}$ (Tutorial 4).
- R transitive $\Leftrightarrow R \circ R \subseteq R$.
- R reflexive $\Rightarrow R \subseteq R \circ R$.
- Asymmetric $\Rightarrow$ antisymmetric (and irreflexive).
- **Antisymmetric is not “not symmetric”**: any $R \subseteq \{(x, x) : x \in A\}$ is both.
- “=” is an equivalence relation **and** a partial order (AY23 Q10, AY24 Q9).
- The empty relation on $A \neq \emptyset$ is symmetric, transitive, antisymmetric and asymmetric, but not reflexive.
- If R, S **both** have the property: reflexivity survives $\cup, \cap, \circ$; symmetry survives $\cup, \cap$ but not $\circ$; transitivity and antisymmetry survive only $\cap$.
- Write “0 is related to itself”, not “0 is reflexive”.

METHOD Proof skeletons for the properties

Reflexive: “Let $x \in A$ so xRx .” **Symmetric**: “Let $x, y \in A$ with xRy so yRx .”

Transitive: “Let $x, y, z \in A$ with xRy and yRz so xRz .” **Antisymmetric**: “Let xRy and yRx so $x = y$.”

Disproof: one explicit counterexample, named elements, each membership checked.

PAST PAPER AY23 Q9 and Q10 · Counterexamples on small sets

Q9 (R reflexive, S symmetric). On $\{a, b, c\}$ take $R = \{(a, a), (b, b), (c, c), (a, c)\}$, $S = \{(a, b), (b, a)\}$: $R \circ S = \{(a, b), (b, a), (b, c)\}$ is not symmetric; $R \cup S$ is not symmetric; $(R \circ S)^{-1}$ is not transitive. Only “ $R \cup S$ is reflexive” survives: **A**.

Q10. Asymmetric $\Rightarrow$ antisymmetric: true. “Not reflexive $\Rightarrow$ not asymmetric”: false ($R = \{(a, b)\}$). “Not reflexive $\Rightarrow$ asymmetric”: false ($R = \{(a, a)\}$ on $\{a, b\}$). “No relation is both an equivalence relation and a partial order”: false (=). Only (i): **A**.

PAST PAPER AY24 Q10, AY25 Q11, AY25 Q14

AY24 Q10. $pRq \Leftrightarrow (p \rightarrow q)$ on $\{T, F\}$: $R = \{(T, T), (F, F), (F, T)\}$ is reflexive, antisymmetric, transitive, not symmetric: **D**.

AY25 Q11. $ARB \Leftrightarrow |A \cap B| = 1$ on $\mathcal{P}(\mathbb{Z})$: symmetric; not reflexive ($|\emptyset \cap \emptyset| = 0$); not transitive. **Symmetric but not an equivalence relation**. Answer B.

AY25 Q14. Reflexive R on $\{1, 2, 3, 4\}$ with $(xRy \wedge xRz) \rightarrow y = z$: reflexivity forces xRx , so each x is related to **nothing else**. Only the identity relation qualifies, so $|\mathcal{F}| = 1$. Answer B.

PROOF R is symmetric $\Leftrightarrow R = R^{-1}$ (Tutorial 4)

1. $(\rightarrow)$ Suppose R symmetric. $(x, y) \in R \rightarrow (y, x) \in R$ (symmetry) $\rightarrow (x, y) \in R^{-1}$ (definition of inverse), so $R \subseteq R^{-1}$. Conversely $(x, y) \in R^{-1} \rightarrow (y, x) \in R \rightarrow (x, y) \in R$ (symmetry), so $R^{-1} \subseteq R$.
2. $(\leftarrow)$ Suppose $R = R^{-1}$ and xRy . Then $(y, x) \in R^{-1} = R$, i.e. yRx . ■

PROOF R is transitive $\Leftrightarrow R \circ R \subseteq R$ (AY23 Q12 model answer, extended)

1. $(\rightarrow)$ Let $(x, z) \in R \circ R$. There is some y with xRy and yRz (definition of composition) so xRz (transitivity), i.e. $(x, z) \in R$.
2. $(\leftarrow)$ Let xRy and yRz . Then $(x, z) \in R \circ R$ (definition of composition) $\subseteq R$, so xRz . ■

Similarly, **reflexive $\Rightarrow R \subseteq R \circ R$** : given $(x, z) \in R$, use xRx and xRz with middle $y = x$.

WORKED EXAMPLE T5 Q7 · Asymmetric vs antisymmetric

On $A = \{a, b, c\}$ find a relation that is (a) asymmetric and antisymmetric; (b) antisymmetric but not asymmetric; (c) asymmetric but not antisymmetric; (d) neither.

(a) $\{(a, b), (a, c)\}$. (b) $\{(a, a), (b, c)\}$. (c) **No solution**: every asymmetric relation is antisymmetric (below). (d) $\{(a, b), (b, a), (a, c)\}$.

1. Let R be an asymmetric relation on A : $\forall x, y \in A (xRy \rightarrow yRx)$ (definition of asymmetry).
 - 1.1. $\equiv \forall x, y \in A (xRy \vee yRx)$ (implication law).
 - 1.2. $\Rightarrow \forall x, y \in A ((xRy \vee yRx) \vee x = y)$ (generalization).
 - 1.3. $\equiv \forall x, y \in A (\sim(xRy \wedge yRx) \vee x = y)$ (De Morgan's law).
 - 1.4. $\equiv \forall x, y \in A (xRy \wedge yRx \rightarrow x = y)$ (implication law).
2. The final formula is the definition of antisymmetry, so R is antisymmetric: asymmetry makes it vacuously true. ■

22 · Closures

DEFINITION Transitive closure (Lecture 6.2.2); reflexive and symmetric closures

The **transitive closure** of R on A is the relation R^t such that (1) R^t is transitive, (2) $R \subseteq R^t$, (3) $R^t \subseteq S$ for every transitive $S \supseteq R$. It is the **smallest** transitive relation containing R . Likewise the **reflexive closure** $R^r = R \cup \{(x, x) : x \in A\}$ and the **symmetric closure** $R^s = R \cup R^{-1}$.

METHOD Computing R^t

$R^t = R \cup R^2 \cup R^3 \cup \dots$ where $R^2 = R \circ R$; on a finite set stop once a power adds nothing new (at most $|A|$ powers). On the digraph: draw $x \rightarrow z$ whenever some path leads from x to z . A cycle back to the starting point creates a loop.

PAST PAPER AY24 Q17(a), (b) (5 marks)

- (a) $R = \{(a, b), (a, c)\}$ on $\{a, b, c\}$: $|R^r| = 2 + 3 = 5$; $|R^s| = |\{(a, b), (a, c), (b, a), (c, a)\}| = 4$; $|R^t| = 2$, since no pair ends where another begins, so R is already transitive.
- (b) $S = \{(a, b), (b, a), (b, d), (d, c), (b, b), (d, d)\}$; $T = S^t \setminus S$ consists of the new pairs obtained from paths: $a \rightarrow b \rightarrow a$ gives (a, a) ; $a \rightarrow b \rightarrow d$ gives (a, d) ; $a \rightarrow b \rightarrow d \rightarrow c$ gives (a, c) ; $b \rightarrow d \rightarrow c$ gives (b, c) . $T = \{(a, a), (a, c), (a, d), (b, c)\}$.

WORKED EXAMPLE T5 Q6 · Reflexive closure (the closure-proof template)

Let R be a relation on a non-empty set A , and define S on A by $xSy \leftrightarrow ((x = y) \vee (xRy))$. Show: (a) S is reflexive; (b) $R \subseteq S$; (c) if S' is a reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is S called?

- (a) Let $x \in A$. $x = x$, so xSx (definition of S). Therefore S is reflexive.
- (b) Suppose $(x, y) \in R$. Then xSy (definition of S), so $(x, y) \in S$. Therefore $R \subseteq S$ (definition of $\subseteq$).
- (c) Let $(x, y) \in S$: $x = y$ or xRy (definition of S).
1. **Case 1** ($x = y$): $xS'y$ (S' is reflexive), so $(x, y) \in S'$.
 2. **Case 2** (xRy): $(x, y) \in R \subseteq S'$, so $(x, y) \in S'$.

In all cases $(x, y) \in S'$. Therefore $S \subseteq S'$. ■

S is the **reflexive closure** of R : the smallest reflexive relation on A containing R . Parts (a)–(c) mirror the three conditions defining a closure.

WORKED EXAMPLE Composition and closure

$A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Find $R \circ R$, $R^{-1} \circ R$ and $|R^t|$.

$R \circ R = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$ (two steps round the 4-cycle). $R^{-1} \circ R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$: follow an arrow of R , then return along the same arrow. From any vertex a path reaches every vertex, itself included, so $R^t = A \times A$ and $|R^t| = 16$.

WORKED EXAMPLE All the closures

$A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3)\}$. Find R^r , R^s , R^t and the smallest equivalence relation containing R .

$R^r = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$ (5 pairs); $R^s = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ (4); $R^t = \{(1, 2), (2, 3), (1, 3)\}$ (3).

The smallest equivalence relation containing R is $A \times A$ (9 pairs): ignoring directions, 1–2–3 is one connected piece, so there is a single class.

THEOREM Closure facts

The transitive closure of a **symmetric** relation is symmetric, and of a **reflexive** relation is reflexive (reverse or extend the paths). Hence the **smallest equivalence relation containing R** is $(R \cup R^{-1} \cup \{(x, x) : x \in A\})^t$. Its classes are the connected components of the digraph of R when arrow directions are ignored.

23 · Equivalence relations

DEFINITION Relation induced by a partition · Equivalence relation

Given a partition $\mathcal{C}$ of A , the **induced relation** is $xRy \leftrightarrow x, y$ lie in the same component; it is reflexive, symmetric and transitive (Theorem 8.3.1).

R on A is an **equivalence relation** iff it is reflexive, symmetric and transitive. Notation: $\sim$.

DEFINITION Equivalence class · Quotient $A/\sim$

For an equivalence relation $\sim$ on A and $a \in A$, the **equivalence class** of a (the **class of a**) is $[a]_{\sim} = \{x \in A : a \sim x\}$; procedurally, $\forall x \in A (x \in [a]_{\sim} \leftrightarrow a \sim x)$.

$A/\sim = \{[x]_{\sim} : x \in A\}$, the set of all equivalence classes, read “the quotient of A by $\sim$ ”. Its **elements** are the equivalence classes.

DEFINITION Congruence modulo n (Lecture 6.3.4)

Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$. a is **congruent to b modulo n** , written $a \equiv b \pmod{n}$, iff $a - b = nk$ for some $k \in \mathbb{Z}$; in other words, $n \mid (a - b)$.

E.g. $7 \equiv 1 \pmod{2}$; $-3 \equiv 12 \pmod{5}$; $-4 \not\equiv 5 \pmod{7}$. The classes are $[x] = \{x + nk : k \in \mathbb{Z}\}$, and $\mathbb{Z}/\sim_n = \{[0], [1], \dots, [n-1]\}$.

PROOF Lemma Rel.1 (Lecture 6)

Let $\sim$ be an equivalence relation on A . For $x, y \in A$, these are equivalent: (i) $x \sim y$; (ii) $[x] = [y]$; (iii)

$[x] \cap [y] \neq \emptyset$.

1. ((i) $\Rightarrow$ (ii)) Suppose $x \sim y$; then $y \sim x$ (symmetry).
 - 1.1. For every $z \in [x]$: $x \sim z$ (definition of $[x]$), so $y \sim z$ (transitivity, as $y \sim x$), so $z \in [y]$. Thus $[x] \subseteq [y]$.
 - 1.2. Symmetrically $[y] \subseteq [x]$, so $[x] = [y]$.
2. ((ii) $\Rightarrow$ (iii)) Suppose $[x] = [y]$. $x \sim x$ (reflexivity), so $x \in [x] = [y]$ (idempotent law) hence $[x] \cap [y] \neq \emptyset$.
3. ((iii) $\Rightarrow$ (i)) Take $z \in [x] \cap [y]$. Then $x \sim z$ and $y \sim z$ (definitions of $[x], [y]$) so $z \sim y$ (symmetry) and $x \sim y$ (transitivity). ■

PROOF Proposition (Lecture 6, slide 54) · Congruence mod n is an equivalence relation on $\mathbb{Z}$ ($a \equiv b \pmod{n} \leftrightarrow n \mid (a - b)$)

1. (Reflexive) For all $a \in \mathbb{Z}$: $a - a = 0 = n \cdot 0$, so $a \equiv a \pmod{n}$ (definition of congruence).
2. (Symmetric) Let $a \equiv b \pmod{n}$: $a - b = nk$ for some $k \in \mathbb{Z}$. Then $b - a = n(-k)$ with $-k \in \mathbb{Z}$ (closure), so $b \equiv a \pmod{n}$.
3. (Transitive) Let $a \equiv b$ and $b \equiv c \pmod{n}$: $a - b = nk$, $b - c = nl$. Then $a - c = n(k + l)$ with $k + l \in \mathbb{Z}$, so $a \equiv c \pmod{n}$. ■

PAST PAPER AY23 Q19 (6 marks) · $aSb \leftrightarrow \exists k \in \mathbb{Z} (a - b = k/2)$ on $\mathbb{Q}$

(a) **Prove S transitive** (official model answer):

1. Let $a, b, c \in \mathbb{Q}$ such that aSb and bSc .
2. There exist $m, n \in \mathbb{Z}$ such that $a - b = m/2$ and $b - c = n/2$ (by definition of S).
3. Now $a - c = (a - b) + (b - c) = m/2 + n/2 = (m + n)/2$ (by basic algebra).
4. $m + n \in \mathbb{Z}$ (by closure of integers under addition).
5. Therefore aSc (by definition of S). ■

(b) Two distinct classes: $[0] = \{k/2 : k \in \mathbb{Z}\}$ and $[0.1]$. (c) $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$. Each class has exactly one representative in $[0, 1/2)$, so there are no duplicates.

PAST PAPER AY25 Q17(c), (d), Q15 · AY24 Q9 · AY23 Q12

AY25 Q17(c). “Same biological father and mother”: **yes**, an equivalence relation. “Share a biological parent”: **no**, because it is not transitive (half-siblings).

AY25 Q17(d). $x \sim y \leftrightarrow \exists d \in \mathbb{Z}^+ (d \mid x \wedge d \mid y)$: $d = 1$ always works, so everything is related: $\mathbb{Z}^+/\sim = \{\mathbb{Z}^+\} = \{[1]\}$.

AY25 Q15. “ $\forall x \in A ([x] \subseteq A/\sim)$ ” is **false**: classes are **elements** of $A/\sim$, not subsets.

“ $\exists x \in A/\sim \forall y \in x (y \in A)$ ” and “ $|A| = \sum_{x \in A/\sim} |x|$ ” are true: **E**.

AY24 Q9. Divisibility on 10 primes is the identity relation: 10 classes **and** 10 minimal elements. Answer **C**.

AY23 Q12. For an equivalence relation R , all four hold: $R = R^{-1}$ (symmetry), so $R^{-1} \circ R = R \circ R^{-1} = R \circ R$; $R \circ R \subseteq R$ (transitivity: $xRy \wedge yRz \Rightarrow xRz$); $R \subseteq R \circ R$ (reflexivity: take the middle $y = x$). Hence $R \circ R^{-1} = R$: **D**.

PROOF Theorem Rel.2 · Theorem 8.3.4 · If $\sim$ is an equivalence relation on A , then $A/\sim$ is a partition of A (Lecture 6)

1. $A/\sim$ is a set (*by definition*).
2. Every element is a non-empty subset of A : $[x] = \{y \in A : x \sim y\} \subseteq A$, and $x \in [x]$ (*reflexivity*).
3. Every $x \in A$ is in exactly one class:
 - 3.1. $x \in [x]$, so x is in at least one class.
 - 3.2. If $x \in [y]$ and $x \in [z]$, then $[y] \cap [z] \neq \emptyset$, so $[y] = [z]$ (*Lemma Rel.1, (iii) $\Rightarrow$ (ii)*). ■

PROOF Theorem 8.3.1 · The relation induced by a partition $\mathcal{C}$ of A is an equivalence relation

1. (Reflexive) Let $x \in A$. x lies in some component S (*definition of partition*), so $x, x \in S$ and xRx .
2. (Symmetric) If xRy , then x, y lie in one component S , so $y, x \in S$ and yRx .
3. (Transitive) Let xRy and yRz : $x, y \in S_1$ and $y, z \in S_2$ for components S_1, S_2 . Then $y \in S_1 \cap S_2$, so $S_1 = S_2$ (*each element lies in exactly one component*), hence $x, z \in S_1$ and xRz . ■

Together with Theorem Rel.2, this gives a one-to-one correspondence between partitions and equivalence relations on A . Counts: §28.

PROOF On $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $(a, b) \sim (c, d) \leftrightarrow ad = bc$ is an equivalence relation

1. (Reflexive) $ab = ba$ (*commutativity*), so $(a, b) \sim (a, b)$.
2. (Symmetric) If $ad = bc$ then $cb = da$ (*commutativity*), so $(c, d) \sim (a, b)$.
3. (Transitive) Let $ad = bc$ and $cf = de$, with $b, d, f \neq 0$.
 - 3.1. Multiply: $adf = bcf = bde$, so $d(af - be) = 0$ (*basic algebra*).
 - 3.2. $d \neq 0$, so $af = be$ (*zero product property*), i.e. $(a, b) \sim (e, f)$. ■

Each class is one rational number: $[(1, 2)] = \{(1, 2), (2, 4), (-3, -6), \dots\}$; $A/\sim$ “is” $\mathbb{Q}$.

METHOD Writing $A/\sim$ without duplicates

Pick **one representative per class** from a region that meets every class exactly once, then write $\{[x] : x \in \text{that region}\}$.

- Congruence mod 3: $\mathbb{Z}/\equiv_3 = \{[0], [1], [2]\}$, with $[r] = \{3k + r : k \in \mathbb{Z}\}$.
- AY23 Q19: $\mathbb{Q}/S = \{[x] : x \in \mathbb{Q} \wedge 0 \leq x < 1/2\}$.
- Lecture Ex. 15: nonempty $A \subseteq \{1, 2, 3\}$, related iff they have the same least element: $\{[1]\} = \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$, $\{[2]\} = \{\{2\}, \{2, 3\}\}$, $\{[3]\} = \{\{3\}\}$, three classes.

PAST PAPER T4 Q8 (AY20/21 exam) · Classes of ordered pairs

On $\mathbb{Z}^+ \times \mathbb{Z}^+$, $(a, b) \sim (c, d) \leftrightarrow ab = cd$. Write the set-builder line first: $[(1, 1)]$

$= \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (1, 1) \sim (x, y)\}$ (*definition of class*)

$= \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 1\}$ (*definition of $\sim$*)

$= \{(1, 1)\}$. Likewise $[(4, 3)] = \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : xy = 12\} = \{(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1)\}$.

WORKED EXAMPLE T4 Q10 · Equivalence relation using Appendix A

On $\mathbb{Z} \setminus \{0\}$ let $a \sim b \leftrightarrow ab > 0$. (a) Prove $\sim$ is an equivalence relation. (b) Determine all distinct equivalence classes.

1. (Reflexivity) Let $a \in \mathbb{Z} \setminus \{0\}$. Since $a \neq 0$, $a^2 > 0$ (*T21*). Thus $a \sim a$ (*definition of $\sim$*).
2. (Symmetry) Let $a, b \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$. Then $ab > 0$ (*definition of $\sim$*), so $ba > 0$ (*commutative law of $\times$*), so $b \sim a$.
3. (Transitivity) Let $a, b, c \in \mathbb{Z} \setminus \{0\}$ with $a \sim b$ and $b \sim c$. Then $ab > 0$ and $bc > 0$ (*definition of $\sim$*).
 - 3.1. Multiplying, $ab^2c > 0$ (*Ord1*), so $(ac)b^2 > 0$ (*associative, commutative laws of $\times$*).
 - 3.2. So ac and b^2 are both positive or both negative (*T25*).
 - 3.3. Since $b^2 > 0$ (*T21, as $b \neq 0$*), $ac > 0$. Thus $a \sim c$.
4. Therefore $\sim$ is an equivalence relation. ■

(b) T25 rules out pairs of opposite signs; the product of two positive or two negative integers is positive. The classes are $[1] = \mathbb{Z}^+$ and $[-1] = \mathbb{Z}^-$, so $(\mathbb{Z} \setminus \{0\})/\sim = \{\mathbb{Z}^+, \mathbb{Z}^-\}$.

DEFINITION Representative · Well-defined operation on $A/\sim$

Any $b \in [a]$ is a **representative** of $[a]$, since then $[b] = [a]$: mod 5, $[2] = [7] = [12]$.

Let $\sim$ be an equivalence relation on A and $*$ a binary operation on A (so $a * b \in A$ for all $a, b \in A$).

The induced rule $[a] * [b] = [a * b]$ is **well defined** iff the answer depends only on the classes, not on the representatives chosen: $a \sim a' \wedge b \sim b' \Rightarrow a * b \sim a' * b'$.

PROOF $+$ and $\cdot$ are well defined on the classes of congruence mod n

1. Fix $n \in \mathbb{Z}^+$ and let $a, a', b, b' \in \mathbb{Z}$ with $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$: $a - a' = nk$ and $b - b' = nm$ for some $k, m \in \mathbb{Z}$ (definition of congruence).
2. (Addition) $(a + b) - (a' + b') = n(k + m)$ with $k + m \in \mathbb{Z}$ (closure), so $a + b \equiv a' + b' \pmod{n}$.
3. (Multiplication) $ab - a'b' = (a - a')b + a'(b - b') = n(kb + a'm)$ with $kb + a'm \in \mathbb{Z}$ (closure), so $ab \equiv a'b' \pmod{n}$. ■

E.g. mod 5: $[2] \cdot [4] = [8] = [3]$ and $[7] \cdot [9] = [63] = [3]$.

TRAP Check reflexive, symmetric, transitive first $\cdot$ then check well-definedness

- $a \sim b \leftrightarrow ab \geq 0$ is reflexive and symmetric, but **not** transitive on either $\mathbb{Z}$ or $\mathbb{R}$ ($1 \sim 0$ and $0 \sim -1$, but $1 \not\sim -1$), so it does not give a partition into equivalence classes. On $\mathbb{Z} \setminus \{0\}$ its classes are $\mathbb{Z}^+$ and $\mathbb{Z}^-$ (§23, T4 Q10). On $\mathbb{R} \setminus \{0\}$ it means “same sign”: $(\mathbb{R} \setminus \{0\})/\sim = \{[1], [-1]\} = \{(0, \infty), (-\infty, 0)\}$.
- $a \sim b \leftrightarrow a - b = 2k$ for some $k \in \mathbb{Z}$, on $\mathbb{R}$: $[a] = \{a + 2k : k \in \mathbb{Z}\}$ and $\mathbb{R}/\sim = \{[r] : 0 \leq r < 2\}$ (infinitely many classes). $+$ is well defined, but $\cdot$ is **not**: $[0.5] = [2.5]$, yet $[0.5] \cdot [0.5] = [0.25] \neq [1.25] = [2.5] \cdot [0.5]$, since $1.25 - 0.25 = 1$ is not of the form $2k$.

WORKED EXAMPLE Equivalence relation and quotient

On $\mathbb{Z}$ let $xRy \leftrightarrow 3 \mid (x^2 - y^2)$. Prove R is an equivalence relation and find $\mathbb{Z}/R$.

1. (Reflexive) $x^2 - x^2 = 0 = 3 \cdot 0$, so xRx (definition of divisibility).
2. (Symmetric) If $x^2 - y^2 = 3k$ then $y^2 - x^2 = 3(-k)$ with $-k \in \mathbb{Z}$, so yRx .
3. (Transitive) If $x^2 - y^2 = 3k$ and $y^2 - z^2 = 3l$, then $x^2 - z^2 = 3(k + l)$, so xRz .
4. (Classes) By the QR theorem $x = 3q + r$ with $r \in \{0, 1, 2\}$; $x^2 = 9q^2 + 6qr + r^2$, so $x^2 \bmod 3 = 0$ if $r = 0$ and 1 if $r = 1, 2$. Hence $\mathbb{Z}/R = \{[0], [1]\} = \{\{3k : k \in \mathbb{Z}\}, \{n \in \mathbb{Z} : 3 \nmid n\}\}$. ■

WORKED EXAMPLE Classes of a relation on pairs

On $\mathbb{Z} \times \mathbb{Z}$ let $(a, b) \sim (c, d) \leftrightarrow a + d = b + c$. Show $\sim$ is an equivalence relation and describe $(\mathbb{Z} \times \mathbb{Z})/\sim$.

1. Rewrite: $(a, b) \sim (c, d) \leftrightarrow a - b = c - d$ (basic algebra). Then reflexivity, symmetry and transitivity are those of $=$ on $\mathbb{Z}$ (e.g. $a - b = c - d$ and $c - d = e - f$ give $a - b = e - f$).
2. So $[(a, b)] = \{(c, d) : c - d = a - b\}$, a diagonal line of lattice points, and each class contains exactly one point $(k, 0)$ with $k = a - b$.
3. $(\mathbb{Z} \times \mathbb{Z})/\sim = \{[(k, 0)] : k \in \mathbb{Z}\}$, with one class for each integer and no duplicates. ■

24 · Partial orders

DEFINITION Partial order (Lecture 6.4.2) · Poset

R on A is a **partial order** iff it is **reflexive, antisymmetric and transitive**; then (A, R) is a **partially ordered set** (poset). A general partial order is written $\preceq$ (“curly less than or equal to”). Examples: $\leq$ on $\mathbb{R}$; $\subseteq$ on any set of sets; $\mid$ on $\mathbb{Z}^+$. Non-examples: $<$ (not reflexive); congruence mod n (not antisymmetric); $\mid$ on $\mathbb{Z}$ ($2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$).

TRAP T5 Q1 · Disprove with one named counterexample

On strings over $\{s, u\}$, $aRb \leftrightarrow \text{len}(a) \leq \text{len}(b)$ is **not** a partial order: sRu and uRs but $s \neq u$, so R is not antisymmetric. One counterexample suffices; there is no need to check reflexivity or transitivity.

PROOF $\mid$ is a partial order on any set $A \subseteq \mathbb{Z}^+$ (Lecture 6, Ex. 20)

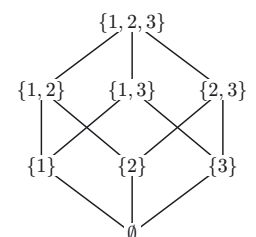
1. (Reflexive) Let $a \in A$. $a = 1 \cdot a$, so $a \mid a$ (by definition of divisibility).
2. (Antisymmetric) Let $a, b \in A$ with $a \mid b$ and $b \mid a$. Then $a \leq b$ and $b \leq a$ (by Theorem 4.3.1, as $a, b \in \mathbb{Z}^+$), so $a = b$.
3. (Transitive) Let $a \mid b$ and $b \mid c$. Then $a \mid c$ (by Theorem 4.3.3). ■

DEFINITION Hasse diagram (Lecture 6.4.3)

For a partial order $\preceq$ on a finite A and distinct x, y : if $x \preceq y$ and no $m \in A$ has $x \preceq m \preceq y$ (with $m \neq x, y$), draw x below y joined by a line; otherwise no line.

From the digraph: place arrows pointing up; delete (1) loops, (2) arrows implied by transitivity, (3) arrowheads.

Reading it: $x \preceq y$ iff you can climb from x to y along lines.



DEFINITION Comparable · Compatible (AY24 paper) · Chain, antichain (AY24 paper) · Total order

a, b are **comparable** iff $a \preccurlyeq b$ or $b \preccurlyeq a$; otherwise **noncomparable**. a, b are **compatible** iff $\exists c \in A (a \preccurlyeq c \wedge b \preccurlyeq c)$.

A **chain** is a subset C in which every two elements are comparable; a **maximal chain** M has no $t \notin M$ with $M \cup \{t\}$ a chain; the **length** of a chain is one less than its number of elements. An **antichain** is a subset in which no two distinct elements are comparable.

$\preccurlyeq$ is a **total (linear) order** iff it is a partial order and $\forall x, y \in A (x \preccurlyeq y \vee y \preccurlyeq x)$; its Hasse diagram is a single chain. $|$ on $\mathbb{Z}^+$ is partial, not total ($3 \nmid 5, 5 \nmid 3$); $\leq$ on $\mathbb{Q}$ is total.

DEFINITION Maximal, minimal, largest, smallest (Lecture 6.4.5)

For a poset $(A, \preccurlyeq)$ and $c \in A$:

- c **maximal** $\Leftrightarrow \forall x \in A (c \preccurlyeq x \rightarrow c = x)$: nothing lies strictly above c . Equivalently, every $x \in A$ has $x \preccurlyeq c$ or is not comparable with c .
- c **minimal** $\Leftrightarrow \forall x \in A (x \preccurlyeq c \rightarrow c = x)$: nothing lies strictly below c . Equivalently, every $x \in A$ has $c \preccurlyeq x$ or is not comparable with c .
- c **largest** $\Leftrightarrow \forall x \in A (x \preccurlyeq c)$: every element lies below c .
- c **smallest** $\Leftrightarrow \forall x \in A (c \preccurlyeq x)$: every element lies above c .

Largest = greatest = maximum; smallest = least = minimum. On a Hasse diagram: maximal = no line going up; minimal = no line going down.

DEFINITION Well-ordered set (Lecture 6.4.7)

Let $\preccurlyeq$ be a total order on a set A . A is **well-ordered** iff every non-empty subset of A contains a smallest element. Symbolically,

$$\forall S \in \mathcal{P}(A) (S \neq \emptyset \rightarrow \exists x \in S \forall y \in S (x \preccurlyeq y)).$$

$(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not: $\mathbb{Z}$ itself (or $\mathbb{Z}^-$) is a non-empty subset with no smallest element.

THEOREM Extremal-element facts

- **Proposition (Lecture 6, slide 83)**: smallest $\Rightarrow$ minimal; largest $\Rightarrow$ maximal. If c is smallest and $x \preccurlyeq c$, then also $c \preccurlyeq x$, so $x = c$ by antisymmetry.
- A largest (smallest) element, if it exists, is **unique** (two largest elements precede each other, hence are equal by antisymmetry), and is then the **only** maximal (minimal) element.
- Conversely, in a **finite** poset a unique maximal element is largest. This fails for infinite posets.
- A finite non-empty poset has at least one maximal and one minimal element.
- Distinct maximal elements are noncomparable (so are distinct minimal ones).
- An element can be both maximal and minimal: an isolated point.
- Infinite posets may have none: $(\mathbb{Z}, \leq)$.
- Every total order on a finite set is well-ordered: a non-empty subset has a minimal element, and in a total order a minimal element is smallest.

PAST PAPER AY23 Q13, Q14 · Facts about partial orders

(i) “No element is both smallest and largest” is false: $(\{a\}, \{(a, a)\})$. (ii) “An element can be maximal and minimal but neither largest nor smallest” is true: $(\{a, b\}, \{(a, a), (b, b)\})$. (iii) “Distinct maximal elements are comparable” is false, as it would contradict maximality. (iv) “There may be no maximal or minimal element” is true: $(\mathbb{Z}, \leq)$. **C.**

Q14. On $\{1, 2, 3\}$: $P = \{(1, 1), (1, 2), (2, 2)\}$ is not reflexive ($(3, 3)$ missing); Q

$= \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 2), (3, 3)\}$ is a partial order; $R = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 1)\}$ is not transitive ($1R2, 2R3$, not $1R3$), and $R^t = A \times A$ is not antisymmetric. Only Q : **B.**

WORKED EXAMPLE Partial order and extremal elements

On $\mathbb{Z}^+$ let $a \preccurlyeq b \Leftrightarrow b = a \cdot 2^k$ for some $k \in \mathbb{N}$. Prove $\preccurlyeq$ is a partial order; find its minimal and maximal elements.

1. (Reflexive) $a = a \cdot 2^0$ and $0 \in \mathbb{N}$.

2. (Antisymmetric) If $b = a2^j$ and $a = b2^k$, then $a = a2^{j+k}$, so $2^{j+k} = 1$, $j = k = 0$, and $a = b$.
3. (Transitive) If $b = a2^j$ and $c = b2^k$ then $c = a2^{j+k}$ with $j + k \in \mathbb{N}$ (closure).
4. (Minimal) $b \preccurlyeq a$ with $b \neq a$ means $a = b2^k$, $k \geq 1$, which is possible iff a is even. So the minimal elements are exactly the **odd** numbers.
5. (Maximal) None: $a \preccurlyeq 2a$ and $2a \neq a$. Hence no largest or smallest element either. ■

25 · Linearization and Kahn's algorithm

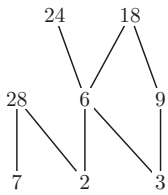
DEFINITION Linearization (Lecture 6.4.7)

A **linearization** of a partial order $\preccurlyeq$ on A is a **total order** $\preccurlyeq^*$ on A with $\forall x, y \in A (x \preccurlyeq y \rightarrow x \preccurlyeq^* y)$. It lists the elements one at a time without breaking any precedence.

METHOD Kahn's algorithm (1962) · Testing a proposed line-up

Kahn: set $A_0 := A$, $i := 0$; while $A_i \neq \emptyset$: pick a **minimal** element c_i of A_i , set $A_{i+1} := A_i \setminus \{c_i\}$, $i := i + 1$. Output $c_0 \preccurlyeq^* c_1 \preccurlyeq^* \dots$. Different choices of minimal element give different linearizations.

Test: a sequence is a linearization iff **for every line of the Hasse diagram, the lower element comes earlier**. A single reversed pair means it is not a linearization.



PAST PAPER AY24 Q13–Q15 · $(\{2, 3, 6, 7, 9, 18, 24, 28\}, |)$

Covers: $2 \mid 6$, $2 \mid 28$, $3 \mid 6$, $3 \mid 9$, $7 \mid 28$, $6 \mid 18$, $6 \mid 24$, $9 \mid 18$.

Q13 Longest maximal chains have length 2:

$\{2, 6, 18\}$, $\{2, 6, 24\}$, $\{3, 6, 18\}$, $\{3, 6, 24\}$, $\{3, 9, 18\}$. There are **5**; answer B.

Q14 $\{2, 3, 28\}$ is not an antichain ($2 \mid 28$); $\{7, 9, 24\}$ is; $\{6\}$ is, vacuously. **D**.

Q15 $2, 3, 6, 7, 9, 18, 24, 28 \checkmark$; $7, 3, 2, 28, 9, 6, 18, 24 \checkmark$; $7, 2, 3, 6, 18, 9, 24, 28 \times$, since $9 \mid 18$ but 18 comes first. **B**.

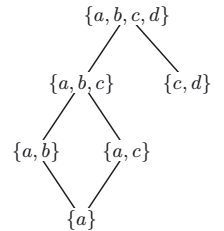
PAST PAPER AY24 Q17(c), (d) · $(E = \{2, 4, 6, 8, 10, 12\}, |)$

Covers: $2 \mid 4$, $2 \mid 6$, $2 \mid 10$, $4 \mid 8$, $4 \mid 12$, $6 \mid 12$. **Minimal:** 2. **Maximal:** 8, 10, 12. **Smallest:** 2. **Largest:** none.

Two maximal chains of different lengths: $\{2, 4, 8\}$ (length 2) and $\{2, 10\}$ (length 1); $\{2, 4, 12\}$, $\{2, 6, 12\}$ also work for the first.

PAST PAPER AY25 Q17(e) · $(\{\{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, \{a, b, c, d\}, \{c, d\}\}, \subseteq)$

Covers: $\{a\} \subset \{a, b\}$, $\{a, c\} \subset \{a, b, c\} \subset \{a, b, c, d\}$, and $\{c, d\} \subset \{a, b, c, d\}$. **Smallest element:** no, since $\{a\}$ and $\{c, d\}$ are both minimal and noncomparable. **Maximal:** yes, $\{a, b, c, d\}$ (which is also largest).



PAST PAPER AY23 Q15, Q18 · $x \preccurlyeq z \leftrightarrow F_x = F_z \vee |F_x| < |F_z|$ on $\{11, \dots, 16\}$, $F_x =$ **positive divisors**

Divisor counts: $11 \rightarrow 2$, $13 \rightarrow 2$, $14 \rightarrow 4$, $15 \rightarrow 4$, $16 \rightarrow 5$, $12 \rightarrow 6$. The order compares counts only, so 11, 13 are incomparable, as are 14, 15. **Minimal:** 11, 13. **Smallest:** none. **Maximal:** 12. **Largest:** 12.

Q15: a linearization must list counts in non-decreasing order: $13, 11, 15, 14, 16, 12 \checkmark$ and $11, 13, 14, 15, 16, 12 \checkmark$; the other two put 12 too early. **B**.

PROOF AY24 Q17(f), (g) · $(a, b)W(c, d) \leftrightarrow a \leq c \wedge b \geq d$ on $\mathbb{Z} \times \mathbb{Z}$

(f) “Every two elements are comparable” is false. Counterexample: $(1, 3)$ and $(2, 4)$, since neither $1 \leq 2 \wedge 3 \geq 4$ nor $2 \leq 1 \wedge 4 \geq 3$ holds.

(g) “Every two elements are compatible” is true.

1. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be arbitrary elements of $\mathbb{Z} \times \mathbb{Z}$.
2. Let $c = (\max(x_1, y_1), \min(x_2, y_2)) \in \mathbb{Z} \times \mathbb{Z}$.
3. Then $x_1 \leq c_1$, $y_1 \leq c_1$ (definition of max) and $x_2 \geq c_2$, $y_2 \geq c_2$ (definition of min).
4. So xWc and yWc (definition of W): x and y are compatible. ■

Relation	Refl.	Sym.	Trans.	Antisym.	Verdict
$=$ on any set	✓	✓	✓	✓	equivalence and partial order
$\leq$ on $\mathbb{R}$	✓	✗	✓	✓	total order
$<$ on $\mathbb{R}$	✗	✗	✓	✓	neither (asymmetric)
$ $ on $\mathbb{Z}^+$	✓	✗	✓	✓	partial order, not total
$ $ on $\mathbb{Z}$	✓	✗	✓	✗	neither ($2 \mid -2 \mid 2$)
$\equiv \pmod{n}$ on $\mathbb{Z}$	✓	✓	✓	✗	equivalence, n classes
$\subseteq$ on $\mathcal{P}(A)$	✓	✗	✓	✓	partial order
$ X = Y $ on $\mathcal{P}(A)$	✓	✓	✓	✗	equivalence, classes by size
$xy \geq 0$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $-1, 0, 1$
$ x - y \leq 1$ on $\mathbb{Z}$	✓	✓	✗	✗	neither: $0, 1, 2$
$\emptyset$ on $A \neq \emptyset$	✗	✓	✓	✓	neither
$A \times A$	✓	✓	✓	✗ if $ A \geq 2$	equivalence, one class

WORKED EXAMPLE Lecture 6, Ex. 26 · Kahn's algorithm on $\{d \in \mathbb{Z}^+ : d \mid 30\}$

Step	Minimal elements of A_i	Choose
A_0	1	$c_0 = 1$
A_1	2, 3, 5	$c_1 = 3$
A_2	2, 5	$c_2 = 2$
A_3	5, 6	$c_3 = 6$
A_4	5	$c_4 = 5$
A_5	10, 15	$c_5 = 15$
A_6	10	$c_6 = 10$
A_7	30	$c_7 = 30$

Result: $1 \prec^* 3 \prec^* 2 \prec^* 6 \prec^* 5 \prec^* 15 \prec^* 10 \prec^* 30$. Any other choice at steps 1–3 or 5 gives another valid linearization.

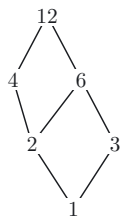
WORKED EXAMPLE Hasse diagram and Kahn

For $(\{1, 2, 3, 4, 6, 12\}, |)$: draw the Hasse diagram; give the minimal, maximal, smallest, largest elements; produce a linearization with Kahn's algorithm.

Covers: $1 \mid 2, 1 \mid 3, 2 \mid 4, 2 \mid 6, 3 \mid 6, 4 \mid 12, 6 \mid 12$ (diagram right). Minimal and smallest: **1**.

Maximal and largest: **12**. Kahn: A_0 has the single minimal element 1; then 2 and 3 are minimal and we take 2; then 3 and 4, and we take 3; then 4 and 6, and we take 4; then 6; then 12.

Linearization: $1 \prec^* 2 \prec^* 3 \prec^* 4 \prec^* 6 \prec^* 12$ (also valid: $1, 3, 2, 6, 4, 12$).



Part VI · Exam Playbook and Locators

26 · Part A: multiple-choice questions

METHOD Approach

- **Judge each of (i)–(iv) on its own** and mark T/F in the margin before reading the options. Several options usually differ in only one item. “None of (A)–(D)” has been the correct answer several times (AY23 Q2, AY24 Q12, AY25 Q2).
- **“For all sets / all relations” claims:** look for a counterexample among $\emptyset$, a singleton, $\{a, b\}$, the empty relation and the identity relation. One counterexample is enough.
- **Relations on small sets:** draw the digraph. **Posets:** draw the Hasse diagram first, then read off the answers.
- **Nested $\emptyset$ / power sets:** write each layer out explicitly, especially for $\mathcal{P}$ of a set containing $\emptyset$.
- **Counting questions:** use the table in §28, or enumerate the $n = 2$ case to test an option.

Before choosing an answer: check quantifier connective and domain (§5–7), membership versus subset (§14), empty sets in products (§15), composition order (§20), antisymmetry (§21), class versus quotient (§23), and maximal versus largest (§24).

27 · Part B: written answers

Computations: give complete set-roster answers; use “None” when appropriate. For a quotient, choose one representative per distinct class. **Proofs:** state the claim, unpack definitions, number and justify each step, then conclude (§9). For “prove or disprove”, state the verdict and supply a proof or one explicit counterexample.

28 · Counting facts

Relations on an n -element set	General	$n = 2$	$n = 3$
all relations	2^{n^2}	16	512
reflexive	2^{n^2-n}	4	64
symmetric	$2^{n(n+1)/2}$	8	64
reflexive and symmetric	$2^{n(n-1)/2}$	2	8
antisymmetric	$2^n \cdot 3^{n(n-1)/2}$	12	216
asymmetric	$3^{n(n-1)/2}$	3	27
equivalence relations (Bell)	B_n	2	5
partial orders	—	3	19
total orders	$n!$	2	6

Derivation. Each of the n^2 pairs is either in the relation or not; reflexivity fixes the n diagonal pairs; symmetry decides each unordered pair once; antisymmetry allows 3 of the 4 states of each off-diagonal pair; asymmetry also forbids loops. For $n = 4$ there are 15 equivalence relations, 219 partial orders and 24 total orders. **AY25 Q4:** on a three-element set there are 5 equivalence relations. **AY25 Q12:** on $\{0, 1\}$ there are 2 equivalence relations and 3 partial orders, so $|F| + 1 = |G|$. Answer B.

Other counts	Value
$ A \times B $	$ A \cdot B $
$ \mathcal{P}(A) $	$2^{ A }$
relations from A to B	$2^{ A \cdot B }$
partitions of an n -set	$B_n : 1, 2, 5, 15, 52$

29 · Question locators

Proof methods: §9–12; set inclusion/equality: §17; relation properties: §21. Each worked example is beside its topic.

Past-paper question	Solved in
AY23 Q2, Q4, Q5, Q16	§6, §4, §7, §7
AY23 Q3, Q6, Q7, Q8, Q17	§2, §18, §16, §14, §18
AY23 Q9, Q10, Q12, Q13, Q14, Q15, Q18, Q19	§21, §21, §23, §24, §24, §25, §25, §23
AY24 Q2, Q3, Q4, Q5	§5, §2, §2, §4
AY24 Q6, Q7, Q8, Q16	§15, §18, §18, §18
AY24 Q9, Q10, Q11, Q12, Q13–15, Q17	§23, §21, §20, §20, §25, §20–§25
AY25 Q1, Q2, Q4, Q6, Q7, Q8, Q9, Q10, Q12	§18, §6, §28, §8, §18, §18, §19, §19, §28
AY25 Q11, Q13, Q14, Q15, Q16, Q17	§21, §4, §21, §23, §4; Q17(a)–(e): §20, §13, §23, §23, §25

Tutorial question	Topic	Solved in
T1 Q11	Critique a biconditional proof; contraposition	§9
T2 Q4	Closure under division; rational addition	§11, §12
T2 Q8, Q11	Cases inside a conditional; factor bound	§11
T3 Q3(c)	Cartesian product versus power set	§15
T3 Q4–6	Double inclusion; element method; set identities	§17
T3 Q7	$A \subseteq B \leftrightarrow A \cup B = B$	§18
T4 Q2	Symmetry and the inverse relation	§21
T4 Q6, Q7, Q9	Repeated composition; associativity; parity relation	§20
T4 Q8, Q10	Classes of pairs; equivalence by sign	§23
T5 Q1	Antisymmetry counterexample on strings	§24
T5 Q6	The three conditions defining a closure	§22
T5 Q7	Asymmetric implies antisymmetric	§21

30 · Self-test: true or false?

Decide each statement before reading the reason.

Statement		Reason
$\sim(p \leftrightarrow q) \equiv \sim p \leftrightarrow q$	T	both are true exactly when p, q differ
$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \vee (q \rightarrow r)$	F	it is $\wedge$; $p = T, q = F, r = F$ separates them
$\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (x < y)$	F	take $x = y$
$\exists x \in \emptyset P(x)$	F	no witness exists; $\forall x \in \emptyset P(x)$ is T
$\emptyset \in \{\emptyset\}$ and $\emptyset \subseteq \{\emptyset\}$	T	listed element; Theorem 6.2.4
$\{\emptyset\} \subseteq \emptyset$	F	$\emptyset \in \{\emptyset\}$ but $\emptyset \notin \emptyset$
$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$	T	$X \subseteq A \cap B \leftrightarrow X \subseteq A \wedge X \subseteq B$
$A \times B = B \times A \Rightarrow A = B$	F	$A = \emptyset, B = \{1\}$: both products are $\emptyset$
a partition of a finite A has at most $ A $ components	T	components are non-empty and disjoint

symmetric and transitive $\Rightarrow$ reflexive	F	the empty relation on $\{a\}$
$R \circ R^{-1}$ is always symmetric	T	$(R \circ R^{-1})^{-1} = (R^{-1})^{-1} \circ R^{-1}$
the inverse of a partial order is a partial order	T	each property survives reversing pairs
the union of two equivalence relations is one	F	$1 \sim 2$ and $2 \sim 3$ from different relations
the intersection of two equivalence relations is one	T	each property is preserved by $\cap$
a unique minimal element is smallest, in any poset	F	true only for finite posets (§24)
$(\mathbb{Z}^+,)$ has a smallest element	T	$1 \mid n$ for all n
there are 4 relations on a one-element set	F	$2^{1^2} = 2$: $\emptyset$ and $\{(a, a)\}$
$a \mid b \wedge a \mid c \Rightarrow a \mid (b - c)$	T	$b - c = a(r - s)$

31 · Symbol table

Symbol	Read as	Symbol	Read as	Symbol	Read as
$\sim p$	not p	$p \wedge q$	p and q	$p \vee q$	p or q
$p \rightarrow q$	if p then q	$p \leftrightarrow q$	p iff q	$p \oplus q$	exclusive or
$P \equiv Q$	logically equivalent	$\therefore$	therefore	$\mathbf{t}, \mathbf{c}$	tautology, contradiction
$\forall$	for all	$\exists$	there exists	$\exists!$	there exists exactly one
$x \in A$	x is an element of A	$A \subseteq B$	A is a subset of B	$A \subsetneq B$	proper subset
$\emptyset$	empty set	$ A $	cardinality	$\mathcal{P}(A)$	power set
$A \times B$	Cartesian product	$A \setminus B$	set difference	$\overline{A}$	complement
$A \oplus B$	symmetric difference	(a, b)	ordered pair	$\{x \in U : P(x)\}$	set-builder
$d \mid n$	d divides n	$d \nmid n$	d does not divide n	$a \equiv b \pmod{n}$	congruent mod n
xRy	$(x, y) \in R$	R^{-1}	inverse relation	$S \circ R$	R then S
R^r, R^s, R^t	closures	$[x]_{\sim}$	class of x	$A/\sim$	quotient: set of classes
$x \preceq y$	x precedes-or-equals y	$x \preceq^* y$	a linearization	$(A, \preceq)$	poset
$\mathbb{Z}^+, \mathbb{Z}_{\geq 0}$	positive / non-negative integers	$\mathbb{N}$	$\{0, 1, 2, \dots\}$	$n \operatorname{div} d, n \operatorname{mod} d$	quotient, remainder

32 · Topic locator

§	Definitions / methods	§	Definitions / methods
1–3	Statements, connectives, equivalence, conditional variants	13–14	Sets, notation, intervals, subset, equality, empty set
4	Arguments, validity, soundness, inference rules	15–16	Pairs, products, set operations, disjointness
5–7	Predicates, truth sets, quantifiers, negation, vacuity	17–18	Set proofs, partitions, power sets
8	Quantified inference rules and worked negations	19–20	Relations, domain/range, inverse, composition
9	Proof vocabulary, format, assumptions, biconditionals	21–22	Relation properties and closures
10	Even/odd, prime, rational, divisibility, absolute value	23	Equivalence, classes, quotient, congruence, representatives
11–12	Proof methods and number-theory examples	24–25	Posets, extrema, total/well orders, Hasse, Kahn
26–29	Exam methods, counting, question locators	30–31	Self-test and symbols
33	Complete Appendix A: field, order, algebra, LUB	34	Slide-numbered theorem locator

Part VII · Appendix A and Theorem Locator

33 · Appendix A · Properties of the Real Numbers

Source: Appendix A, cited in Lecture 1, slide 26; Lecture 3, slide 81; and Lecture 4, slide 4. The complete appendix is reproduced here as a statement reference, including its final least upper bound axiom. The symbols a, b, c, d below denote arbitrary **real numbers**, subject to each stated condition.

Appendix A · Field axioms F1–F6

Addition and multiplication are binary operations on $\mathbb{R}$: if $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$.

Appendix A label	Statement
Axiom F1 · Commutative laws	$a + b = b + a$; $ab = ba$.
Axiom F2 · Associative laws	$(a + b) + c = a + (b + c)$; $(ab)c = a(bc)$.
Axiom F3 · Distributive laws	$a(b + c) = ab + ac$; $(b + c)a = ba + ca$.
Axiom F4 · Identity elements	There are distinct $0, 1 \in \mathbb{R}$ such that $0 + a = a + 0 = a$ and $1 \cdot a = a \cdot 1 = a$.
Axiom F5 · Additive inverses	For each a , there is $-a \in \mathbb{R}$ with $a + (-a) = (-a) + a = 0$.
Axiom F6 · Reciprocals	For each $a \neq 0$, there is $a^{-1} = 1/a \in \mathbb{R}$ with $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Appendix A · Theorems T1–T16 (algebra)

Appendix A label	Statement
Theorem T1 · Addition cancellation	If $a + b = a + c$, then $b = c$. The additive identity 0 is unique.
Theorem T2 · Subtraction	For each a, b , exactly one x satisfies $a + x = b$. This x is denoted $b - a$; $0 - a = -a$.
Theorem T3	$b - a = b + (-a)$.
Theorem T4	$-(-a) = a$.
Theorem T5	$a(b - c) = ab - ac$.
Theorem T6	$0 \cdot a = a \cdot 0 = 0$.
Theorem T7 · Multiplication cancellation	If $ab = ac$ and $a \neq 0$, then $b = c$. The multiplicative identity 1 is unique.
Theorem T8 · Division	For $a \neq 0$, exactly one x satisfies $ax = b$. This x is denoted b/a ; $1/a$ is the reciprocal of a .
Theorem T9	If $a \neq 0$, then $b/a = b \cdot a^{-1}$.
Theorem T10	If $a \neq 0$, then $(a^{-1})^{-1} = a$.
Theorem T11 · Zero product	If $ab = 0$, then $a = 0$ or $b = 0$.
Theorem T12 · Negative signs	$(-a)b = a(-b) = -(ab)$; $(-a)(-b) = ab$. If $b \neq 0$, $-(a/b) = (-a)/b = a/(-b)$.
Theorem T13 · Equivalent fractions	If $b \neq 0$ and $c \neq 0$, then $a/b = (ac)/(bc)$.
Theorem T14 · Addition of fractions	If $b \neq 0$ and $d \neq 0$, then $a/b + c/d = (ad + bc)/(bd)$.
Theorem T15 · Multiplication of fractions	If $b \neq 0$ and $d \neq 0$, then $(a/b)(c/d) = (ac)/(bd)$.
Theorem T16 · Division of fractions	If $b \neq 0$, $c \neq 0$ and $d \neq 0$, then $(a/b)/(c/d) = (ad)/(bc)$.

Appendix A · Order axioms Ord1–Ord3

Appendix A label	Statement
Axiom Ord1	If a and b are positive, then $a + b$ and ab are positive.
Axiom Ord2	If $a \neq 0$, exactly one of a and $-a$ is positive.
Axiom Ord3	0 is not positive.

Appendix A · Order definitions: $a < b$ means $b + (-a)$ is positive; $b > a$ means $a < b$. Also, $a \leq b$ means $a < b$ or $a = b$, and $b \geq a$ means $a \leq b$. A number a is **negative** if $a < 0$, and **nonnegative** if $a \geq 0$.

Appendix A · Theorems T17–T27 (order)

Appendix A label	Statement
Theorem T17 · Trichotomy	Exactly one holds: $a < b$, $b < a$, or $a = b$.
Theorem T18 · Transitivity	If $a < b$ and $b < c$, then $a < c$.
Theorem T19	If $a < b$, then $a + c < b + c$.
Theorem T20	If $a < b$ and $c > 0$, then $ac < bc$.
Theorem T21	If $a \neq 0$, then $a^2 > 0$.
Theorem T22	$1 > 0$.
Theorem T23	If $a < b$ and $c < 0$, then $ac > bc$.
Theorem T24	If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.
Theorem T25	If $ab > 0$, then a and b are both positive or both negative.
Theorem T26	If $a < c$ and $b < d$, then $a + b < c + d$.
Theorem T27	If $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$.

AXIOM Appendix A · Axiom LUB · Least upper bound

Every nonempty subset S of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$. Explicitly, let $B = \{u \in \mathbb{R} : \forall s \in S, s \leq u\}$.

If $S \neq \emptyset$ and $B \neq \emptyset$, then there is $\ell \in B$ such that $\forall u \in B, \ell \leq u$. This ℓ is the **least upper bound** (supremum) of S . This axiom holds for $\mathbb{R}$; the analogous statement for subsets and bounds in $\mathbb{Q}$ fails.

Lecture 4, slide 4 · Additional assumptions: equality is reflexive, symmetric and transitive for all objects; no integer lies strictly between 0 and 1; and $\mathbb{Z}$ is closed under addition, subtraction and multiplication. Lecture 1 additionally permits the even/odd dichotomy (Assumption 1, slide 27) and reduction of every rational to lowest terms (Assumption 2, slide 37).

34 · Theorem locator · Slide numbering

Theorems are stated with their proofs or definitions in the listed sections. Labels match the slides. **L** = lecture; **s** = slide.

Slide label	Topic	Source	Section
Theorem 2.1.1	Logical equivalences	L2 s25–26	§2
Theorem 3.2.1; Theorem 3.2.2	Negating $\forall$; negating $\exists$	L3 s29–30	§6
Theorem 4.2.1	Integers are rational	L4 s19	§12
Theorem 4.2.2; Corollary 4.2.3	Sum; double of a rational	L4 s20–21	§12
Theorem 4.3.1	Positive divisor bound	L4 s24	§12
Theorem 4.3.2	Divisors of 1	L4 s25	§12
Theorem 4.3.3	Transitivity of divisibility	L4 s26	§12
Theorem 4.4.1	Quotient–remainder	L5 s32	§10
Lemma 4.4.4; Theorem 4.4.6	Absolute value bounds; triangle inequality	L1 s24	§12
Theorem 4.6.1	No greatest integer	L4 s29	§12
Proposition 4.6.4	Even square implies even integer	L1 s38; L4 s31–32	§9
Theorem 4.7.1	Irrationality of $\sqrt{2}$	L1 s36, 39; L4 s28	§12
Theorem 6.2.1	Subset relations	L5 s40	§16
Theorem 6.2.2	Set identities	L5 s42–43	§2
Theorem 6.2.4	Empty set	L5 s15	§14
Theorem 6.3.1	Power-set cardinality	L5 s34–36	§18
Theorem 8.3.1	Relation induced by a partition	L6 s39	§23
Lemma Rel.1	Equal or disjoint classes	L6 s47–49	§23
Theorem 8.3.4; Theorem Rel.2	Classes partition the underlying set	L6 s50, 57–59	§23
Propositions (unnumbered)	Composition: associativity; inverse	L6 s18	§20
Proposition (unnumbered)	Congruence is an equivalence relation	L6 s54	§23
Proposition (unnumbered)	Smallest implies minimal	L6 s83	§24